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PROFILE-STABLE BUFFERED MULTIPLICITY FACTORING IN FOUR DIMENSIONS

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Abstract. Multiplicity factoring is usually formulated for child families at comparable scales. For children of mixed geometry, thickening at the shortest parent scale produces nonuniform inflation ratios, and a single worst-case replacement does not preserve the natural density normalization.

We prove a multiplicity-factoring theorem for finite indexed convex parent–child families in $\mathbb{R}^4$ that accommodates arbitrary child shapes, scales, orientations, aspect ratios, and repetitions. The local geometry of each assigned family is encoded by a thickening-weighted Frostman coefficient and a mean-normalized inflation efficiency, both determined by the base family before any shading refinement. The resulting coarse density satisfies

$$\mathfrak{L}^{-A_\varepsilon}(w_1/w_4)^{\varepsilon}\mathfrak{P}_\varepsilon\lambda^{K_\varepsilon},$$

up to the stated parameter-dependent constant, where $\mathfrak{P}_\varepsilon$ is an explicit profile of the parent loads and local efficiencies.

The proof thickens arbitrary measurable shadings, projects along a shortest parent direction, establishes an indexed three-dimensional convex-union estimate in the presence of collisions, and lifts the resulting density response back to four dimensions. A weighted Hölder inequality then assembles the nonuniform parent data, while a common cellular refinement regularizes the fine and coarse multiplicities and yields the stated parentwise multiplicity-product estimate.

Under a relative convex Frostman hypothesis, the profile is expressed explicitly in the minimum, mean, and maximum inflation ratios. The comparable-scale regime follows as a specialization after a preliminary load selection. Thus the theorem supplies a structural factoring input for four-dimensional overlap arguments; deriving new Kakeya maximal or Hausdorff-dimension estimates would require additional analytic ingredients.

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1. Introduction

Multiplicity factoring separates two sources of overlap in a parent–child family: the overlap among the parents and the overlap among the children assigned to one parent. This separation is a recurring feature of Kakeya arguments, from the overlap methods of Córdoba and Wolff through Katz–Łaba–Tao, Katz–Tao, and the multilinear polynomial method [3, 12, 8, 9, 4]; see also [13]. The convex-set framework of Wang and Zahl and the

factoring perspective of Guth, Wang, and Zahl provide the three-dimensional background for the argument below [10, 5].

A comparable-scale formulation makes this factorization particularly transparent. When the children have uniformly comparable John axes, thickening them at a fixed radius changes their volumes by comparable factors, and a single local density can be passed through the argument. That normalization is too rigid once the assigned children have genuinely different geometries. At the shortest parent scale h, the quantities

$$r_W(V) = \frac{|N_h(V)|}{|V|}$$

may vary widely, even within one parent. Replacing them by the largest inflation ratio loses the mean information needed to compare the original and thickened densities, while discarding the weights breaks the relative Frostman normalization.

The phenomenon is not limited to children of different volumes. If $\tau \ll h$, consider equal-volume bodies with John-axis quadruples

$$(\tau^3, \tau, \tau, \tau) \quad \text{and} \quad (\tau^2, \tau^2, \tau, \tau).$$

Their coordinatewise axis ratios become unbounded as $\tau \to 0$, but both h-neighborhoods have volume comparable to h^4. Thus their inflation ratios are comparable although the usual coordinatewise hypothesis fails. Example 7.2 gives the precise calculation. This model suggests that the relevant local datum is the behavior under thickening, rather than a direct comparison of the child axes.

We implement this idea by weighting each child with its inflation ratio. For an assigned family, the minimum, volume-weighted mean, and maximum inflation ratios are denoted by

$$r_W^- \le \overline{r}_W \le r_W^+.$$

The thickening-weighted relative Frostman coefficient $C^{\mathrm{wt}}_{F,h}(\mathcal{V}_W, W)$ measures how the weighted body mass can concentrate in a convex subregion of the parent. Together with the mean-normalized density conversion it gives the local efficiency κ_W. Under an ordinary relative Frostman hypothesis, this efficiency is bounded below by the explicit quantity

$$\eta_W^{(K)} = \frac{\overline{r}_W}{r_W^+} \left(\frac{r_W^-}{\overline{r}_W} \right)^K.$$

All of these are base-family data: they are determined before the shading is refined and remain fixed throughout the construction.

Theorem 2.2 applies this local normalization to finite indexed convex parent–child families in $\mathbb{R}^4$. The parents have comparable John dimensions, but the children may have arbitrary shapes, scales, aspect ratios, orientations, and indexed repetitions. At the common analytic radius $h = w_1/100$, each parent is replaced by the stable buffer $W^\sharp = N_{w_1}(W)$. A single refinement then regularizes the inner, global, and coarse multiplicities, satisfies the

parentwise multiplicity product, and produces compatible fine and coarse fibres. The density of the coarse shading has the form

$$\mathfrak{L}^{-A_\varepsilon}\left(\frac{w_1}{w_4}\right)^{\varepsilon}\mathfrak{P}_\varepsilon\lambda^{K_\varepsilon},$$

up to the stated dependence on ε and the parent comparability constant. Here $\mathfrak{P}_\varepsilon$ is an explicit profile of the local efficiencies and parent loads.

The load profile is needed because the main theorem keeps every parent with positive initial shaded mass. If $D_W = T_W/|W|$ denotes the child load of a parent, the weighted Hölder coefficient

$$\mathfrak{P}_K(\mathbf{D},\boldsymbol{\alpha};\mathbf{p}) = \frac{(\sum_W p_W D_W)^K}{(\sum_W p_W D_W^{K/(K-1)}\alpha_W^{-1/(K-1)})^{K-1}}$$

assembles the nonuniform local estimates without first selecting a common load level. It records the actual obstruction created by the fixed coarse denominator; when the loads and local efficiencies are balanced, it reduces to a positive constant of the expected size.

The local argument has three geometric components. First, arbitrary measurable shadings are thickened with their exact inflation weights. The resulting bodies are projected along a shortest John direction of the parent, which makes the volume of each thickened body comparable to h times its projected volume without restricting the orientations of the children. The projected family need not be essentially distinct: different indexed children may produce the same three-dimensional body. We therefore pass from the essentially-distinct convex-union estimate of Wang and Zahl to an indexed estimate by means of affine-adapted canonical boxes. On each collision layer, the loss in geometric body mass is canceled by the gain in indexed Katz–Tao density. Finally, a layer-cake and Jensen argument lifts the three-dimensional density response back to four dimensions without changing its exponent.

The global argument combines these parentwise estimates through one cellular refinement. Two initial selections make the inner and global fine multiplicities uniform. A global null cleanup aligns pointwise membership with positive-measure cell activity. The remaining selections regularize the indexed active-parent count and the fine mass of a cell. Because every stage restricts all children by a common union of half-open cells, the multiplicity conclusions, fibre relation, local union uniformity, and density estimate hold for the same refinement. The weighted Hölder inequality then converts the sum of the local responses into the displayed profile density.

Comparable child geometry reappears as a specialization rather than as an input to the main theorem. Corollary 7.3 first selects a dyadic parent-load level; coordinatewise child-axis comparability then makes the inflation efficiencies uniformly positive. The general profile consequently collapses to the familiar comparable-scale conclusion, with the corresponding selected coarse normalization. The distinction between the two normalizations is essential when the original loads are highly unbalanced.

The result is intended as a structural factoring theorem for four-dimensional overlap arguments. It does not by itself supply the restriction, two-ends, transversality, or other analytic input needed for a new four-dimensional Kakeya maximal or Hausdorff-dimension estimate.

Section 2 introduces the indexed conventions and profile data and states the main results. Section 3 develops the codimension-one lift. Section 4 proves the indexed three-dimensional convex-union estimate, and Section 5 converts it into the nonuniform local theorem in four dimensions. Section 6 contains the Hölder assembly and cellular construction. Section 7 treats balanced profiles and the comparable-scale regime, while Section 8 explains the geometric and normalization choices through model examples. The appendices collect quantitative dependence, the derivation of the external input, and supplementary boundary cases.

2. Indexed families and the profile theorem

This section fixes the indexed conventions and the quantities used to describe the geometry inside one parent. The order is important: the body family determines the inflation and concentration data first, and subsequent shading refinements are measured against those fixed denominators. We then state the main theorem in a form that keeps the initial positive-mass parent support visible in the coarse normalization.

2.1. **Convex bodies and indexed notation.** A convex body in $\mathbb{R}^n$ is compact, convex, nonempty, and has positive n-dimensional Lebesgue measure. If K is a convex body, its John ellipsoid is the maximal-volume ellipsoid $E_K \subset K$ [7]. We normalize it by

$$E_K \subset K \subset nE_K. \tag{1}$$

Its ordered semiaxis lengths are $a_1(K) \le \cdots \le a_n(K)$. A John box is aligned with the principal axes and has side lengths comparable to these semiaxes.

All families are finite and indexed. Thus $\mathcal{A} = (A_i)_{i\in I}$ may contain $A_i = A_j$ for $i \ne j$, and sums and multiplicities count indices. For a shading $Y(A_i) \subset A_i$, write

$$U(\mathcal{A}, Y) = \bigcup_{i\in I} Y(A_i), \qquad M(\mathcal{A}, Y) = \sum_{i\in I} |Y(A_i)|,$$

$$T(\mathcal{A}) = \sum_{i\in I} |A_i|, \qquad \lambda(\mathcal{A}, Y) = \frac{M(\mathcal{A}, Y)}{T(\mathcal{A})},$$

$$m_{\mathcal{A},Y}(x) = \sum_{i\in I} \mathbf{1}_{Y(A_i)}(x), \qquad \mu(\mathcal{A}, Y) = \frac{M(\mathcal{A}, Y)}{|U(\mathcal{A}, Y)|}.$$

The last quantity is used only for positive-measure unions. The fibre at x is

$$(\mathcal{A})_Y(x) = \{i \in I : x \in Y(A_i)\}.$$

A refinement satisfies $Y'(A_i) \subset Y(A_i)$ for every index. The base family continues to carry all of its indices, with empty refined shading on a discarded

index, except when a theorem first restricts to its positive-mass input support and then explicitly extends the output back to the original family.

For a positive-volume convex test body K, define

$$\Delta(\mathcal{A}, K) = \frac{1}{|K|} \sum_{i: A_i \subset K} |A_i|, \qquad \Delta_{\max}(\mathcal{A}) = \sup_K \Delta(\mathcal{A}, K).$$

If every $A_i \subset W$, the relative Frostman constant is

$$C_F(\mathcal{A}, W) = \sup_{\substack{K \subset W \\ K \text{ a convex body}}} \frac{\Delta(\mathcal{A}, K)}{\Delta(\mathcal{A}, W)}. \tag{2}$$

This is imposed only on nonempty families. For $h > 0$, let $N_h(A) = \{x : \operatorname{dist}(x, A) \le h\}$, and put $\log_+ t = \max\{0, \log t\}$.

2.2. **Local profile data determined by the base family.** Let $\mathcal{V}_W$ be a nonempty assigned indexed family of convex bodies $V \subset W$. Thickening at radius h changes the volume of each child by a factor that depends on its shape and scale. The quantities below retain the minimum, maximum, and body-mass-weighted mean of these inflation factors, while D_W records the total child load in the parent. Set

$$T_W = \sum_{V \in \mathcal{V}_W} |V|, \qquad D_W = \frac{T_W}{|W|}, \qquad r_W(V) = \frac{|N_h(V)|}{|V|},$$

$$r_W^- = \min_{V \in \mathcal{V}_W} r_W(V), \qquad r_W^+ = \max_{V \in \mathcal{V}_W} r_W(V), \qquad \overline{r}_W = \frac{\sum_V r_W(V)|V|}{T_W}. \tag{3}$$

Thus $0 < r_W^- \le \overline{r}_W \le r_W^+ < \infty$. For any measurable shading or later refinement Y, define

$$d_W(Y) = \frac{\sum_V |Y(V)|}{T_W}, \qquad d_{W,h}^{\mathrm{wt}}(Y) = \frac{\sum_V r_W(V)|Y(V)|}{\sum_V r_W(V)|V|}. \tag{4}$$

Since $r_W(V) \ge r_W^-$,

$$d_{W,h}^{\mathrm{wt}}(Y) \ge \frac{r_W^-}{\overline{r}_W}\, d_W(Y). \tag{5}$$

The denominator in the weighted density is the total thickened body mass. This is why the conversion in (5) involves the mean $\overline{r}_W$ rather than the worst inflation r_W^+.

The same weights also alter the relative Frostman distribution. The next coefficient measures how much of the weighted body mass can concentrate in a convex subregion of W, normalized by its density in the full parent.

Definition 2.1 (Weighted relative Frostman coefficient)**.** Define

$$C^{\mathrm{wt}}_{F,h}(\mathcal{V}_W, W) = \sup_{\substack{K\subset W\\ K \text{ a convex body}}} \frac{|K|^{-1}\sum_{\substack{V\in\mathcal{V}_W\\ V\subset K}} r_W(V)|V|}{|W|^{-1}\sum_{V\in\mathcal{V}_W} r_W(V)|V|}. \tag{6}$$

Testing $K = W$ shows $C^{\mathrm{wt}}_{F,h} \geq 1$. This coefficient and all quantities in (3) depend only on the base family, not on a shading. The two efficiencies below combine weighted concentration with the density conversion. The first retains the exact weighted Frostman coefficient; the second is its explicit counterpart under an ordinary relative Frostman hypothesis. For $K > 1$, put

$$\kappa^{(K)}_W = \left(C^{\mathrm{wt}}_{F,h}(\mathcal{V}_W, W)\right)^{-1} \left(\frac{r^-_W}{\overline{r}_W}\right)^K, \tag{7}$$

$$\eta^{(K)}_W = \frac{\overline{r}_W}{r^+_W}\left(\frac{r^-_W}{\overline{r}_W}\right)^K = \frac{(r^-_W)^K}{r^+_W(\overline{r}_W)^{K-1}}. \tag{8}$$

Both numbers lie in $(0, 1]$.

The local efficiencies must eventually be combined across parents whose loads may be very different. For positive D_W, α_W and probability weights p_W, define

$$\mathfrak{P}_K(\mathbf{D}, \boldsymbol{\alpha}; \mathbf{p}) = \frac{\left(\sum_W p_W D_W\right)^K}{\left(\sum_W p_W D_W^{K/(K-1)} \alpha_W^{-1/(K-1)}\right)^{K-1}}. \tag{9}$$

2.3. **The four-dimensional profile theorem.** Let

$$\mathcal{V} = \dot{\bigcup}_{W\in\mathcal{W}} \mathcal{V}_W$$

be a disjoint union by indices. Repetitions are not identified. Given the input shading, define its initial positive-mass parent support

$$\mathcal{W}_0 = \left\{ W \in \mathcal{W} : M^{(0)}_W := \sum_{V\in\mathcal{V}_W} |Y(V)| > 0 \right\}. \tag{10}$$

When the global shaded mass is positive, $\mathcal{W}_0$ is nonempty. All profile sums and the frozen coarse denominator below are over $\mathcal{W}_0$. Removing the other parent groups preserves shaded mass and can only increase the fine density. We denote the indexed buffer family $(W^\sharp)_{W\in\mathcal{W}_0}$ by $\mathcal{W}_0^\sharp$.

Theorem 2.2 (Profile-stable buffered factoring in $\mathbb{R}^4$)**.** *Fix $\Lambda_W \geq 1$ and $0 < w_1 \leq w_2 \leq w_3 \leq w_4$. Suppose that:*

(H1) *$V \subset W$ whenever $V \in \mathcal{V}_W$;*

(H2) *every parent satisfies*

$$\Lambda_W^{-1} w_j \leq a_j(W) \leq \Lambda_W w_j, \qquad 1 \leq j \leq 4;$$

(H3) *the $Y(V)\subset V$ are measurable and*

$$\lambda=\frac{\sum_{V\in\mathcal{V}}|Y(V)|}{\sum_{V\in\mathcal{V}}|V|}>0.$$

There is no comparability hypothesis on child axes, volumes, aspect ratios, or orientations. Set $h=w_1/100$, $W^\sharp=N_{w_1}(W)$, and

$$\mathfrak{L}=2+\log(2+\#\mathcal{V})+\log(2+\#\mathcal{W})+\log(2+w_4/w_1).$$

For every $\varepsilon>0$ there are $K_\varepsilon\ge 2$ and $A_\varepsilon<\infty$ such that the following holds. For $W\in\mathcal{W}_0$, form the frozen data above at radius h, put

$$p_W=\frac{|W|}{\sum_{W'\in\mathcal{W}_0}|W'|},\qquad q_\varepsilon=\frac{K_\varepsilon}{K_\varepsilon-1},$$

and define

$$\mathfrak{P}_\varepsilon^{\mathrm{int}}=\frac{\left(\sum_{W\in\mathcal{W}_0}p_W D_W\right)^{K_\varepsilon}}{\left(\sum_{W\in\mathcal{W}_0}p_W D_W^{q_\varepsilon}(\kappa_W^{(K_\varepsilon)})^{-1/(K_\varepsilon-1)}\right)^{K_\varepsilon-1}}. \tag{11}$$

Then there exist one refinement Y', coarse shadings $Z(W^\sharp)\subset W^\sharp$ for every $W\in\mathcal{W}_0$, a nonempty final positive-mass set

$$\mathcal{W}_+=\left\{W\in\mathcal{W}_0:M'_W:=\sum_{V\in\mathcal{V}_W}|Y'(V)|>0\right\},$$

positive dyadic a,b,c, and $\sigma>0$, with $Z(W^\sharp)=\varnothing$ for $W\notin\mathcal{W}_+$, such that

$$\sum_V|Y'(V)|\gtrsim\mathfrak{L}^{-A_\varepsilon}\sum_V|Y(V)|, \tag{12}$$

$$U'_W\subset Z(W^\sharp)\subset N_{w_1}(U'_W)\cap W^\sharp\quad(W\in\mathcal{W}_+), \tag{13}$$

$$\lambda(\mathcal{W}_0^\sharp,Z)\gtrsim_{\varepsilon,\Lambda_W}\mathfrak{L}^{-A_\varepsilon}\left(\frac{w_1}{w_4}\right)^{\varepsilon}\mathfrak{P}_\varepsilon^{\mathrm{int}}\lambda^{K_\varepsilon}, \tag{14}$$

$$a\le m_{\mathcal{V}_W,Y'}(x)<2a\quad(x\in U'_W,\ W\in\mathcal{W}_+), \tag{15}$$

$$c\le m_{\mathcal{V},Y'}(x)<2c\quad(x\in U(\mathcal{V},Y')), \tag{16}$$

$$b\le m_{\mathcal{W}_0^\sharp,Z}(x)<2b\quad(x\in U(\mathcal{W}_0^\sharp,Z)). \tag{17}$$

Here $U'_W=\bigcup_{V\in\mathcal{V}_W}Y'(V)$. For every $W\in\mathcal{W}_+$,

$$\mu(\mathcal{V},Y)\lesssim_{\varepsilon,\Lambda_W}\mathfrak{L}^{A_\varepsilon}\mu(\mathcal{W}_0^\sharp,Z)\mu(\mathcal{V}_W,Y'). \tag{18}$$

For every $x\in\mathbb{R}^4$,

$$(\mathcal{V})_{Y'}(x)=\dot{\bigcup_{\substack{W\in\mathcal{W}_0\\ x\in Z(W^\sharp)}}}(\mathcal{V}_W)_{Y'}(x). \tag{19}$$

Moreover,

$$|U(\mathcal{V},Y')\cap B(x,w_1)|\asymp\sigma\quad(x\in U(\mathcal{W}_0^\sharp,Z)), \tag{20}$$

$$\Delta_{\max}(\mathcal{W}_0^\sharp)\lesssim_{\Lambda_W}\Delta_{\max}(\mathcal{W}). \tag{21}$$

Coarse normalization. The set $\mathcal{W}_0$ is fixed by the input shading, whereas $\mathcal{W}_+$ records the parents that retain positive mass after the refinement. The density denominator remains $\sum_{W\in\mathcal{W}_0}|W^\sharp|$; parents in $\mathcal{W}_0\setminus\mathcal{W}_+$ carry empty coarse shading but stay in that denominator. Moreover, $Z(W^\sharp)$ is a union of selected cells, so a point may belong to a coarse shading even when its fine fibre for that parent is empty. The fibre identity is formulated accordingly.

Corollary 2.3 (Elementary C-Frostman profile)**.** *In Theorem 2.2, suppose additionally that every $\mathcal{V}_W$, $W\in\mathcal{W}_0$, is C-Frostman in W for one $C\ge 1$. Define*

$$\mathfrak{P}_\varepsilon^{\mathrm{elem}}=\frac{\left(\sum_{W\in\mathcal{W}_0}p_W D_W\right)^{K_\varepsilon}}{\left(\sum_{W\in\mathcal{W}_0}p_W D_W^{q_\varepsilon}(\eta_W^{(K_\varepsilon)})^{-1/(K_\varepsilon-1)}\right)^{K_\varepsilon-1}}. \tag{22}$$

The same objects may be chosen so that

$$\lambda(\mathcal{W}_0^\sharp,Z)\gtrsim_{\varepsilon,\Lambda_W}\mathfrak{L}^{-A_\varepsilon}\left(\frac{w_1}{w_4}\right)^{\varepsilon}C^{-1}\mathfrak{P}_\varepsilon^{\mathrm{elem}}\lambda^{K_\varepsilon}. \tag{23}$$

The indexed formulation permits arbitrary repetitions. Their contributions remain visible in the cardinalities entering $\mathfrak{L}$, in the multiplicities, and in the displayed profile data, while the implicit constants require no separate bound on duplicate multiplicity. Likewise, the child minimum scales and child axis ratios enter through the explicit inflation and profile quantities rather than through $\mathfrak{L}$ or an implicit constant.

3. A codimension-one density-response lift

The local four-dimensional estimate will be obtained from a three-dimensional union bound after projection. The following measure-theoretic lemma returns the projected response to the original dimension. It uses only layer cake and convexity of the response function, so the density exponent is preserved without introducing a new geometric loss.

Theorem 3.1 (Density-response lift)**.** *Let $I\subset\mathbb{R}$ be an interval of length $H>0$, and let $\mathcal{Q}=(Q_i)_{i\in J}$ be a nonempty finite indexed family of measurable subsets of $\mathbb{R}^d$ with*

$$D:=\sum_{i\in J}|Q_i|\in(0,\infty).$$

Suppose that for every measurable $E_i\subset Q_i$ one has

$$|\bigcup_i E_i|\ge A(\mathcal{Q})\,\Psi\left(\frac{\sum_i|E_i|}{D}\right), \tag{24}$$

where $A(\mathcal{Q})\ge 0$ is finite and independent of the shadings, $\Psi:[0,1]\to[0,\infty)$ is convex, and $\Psi(0)=0$. Then, for every measurable $S_i\subset I\times Q_i$,

$$|\bigcup_i S_i|\ge HA(\mathcal{Q})\,\Psi\left(\frac{\sum_i|S_i|}{HD}\right). \tag{25}$$

Proof. For almost every $z \in \mathbb{R}^d$, define the one-dimensional fibre and its length by

$$(S_i)_z = \{s \in I : (s, z) \in S_i\}, \qquad f_i(z) = |(S_i)_z|.$$

Tonelli's theorem gives measurable representatives with $0 \le f_i \le H$, with $f_i = 0$ almost everywhere off Q_i, and

$$\int_{\mathbb{R}^d} f_i(z)\, dz = |S_i|. \tag{26}$$

The fibre of the finite union contains each individual fibre, so

$$|(\bigcup_i S_i)_z| \ge \max_i f_i(z)$$

for almost every z. Put

$$E_i(t) = \{z : f_i(z) > t\}, \qquad \rho(t) = \frac{\sum_i |E_i(t)|}{D}, \qquad 0 < t < H.$$

The sets $E_i(t)$ are measurable subsets of Q_i, and layer cake gives

$$|\bigcup_i S_i| \ge \int_{\mathbb{R}^d} \max_i f_i(z)\, dz \tag{27}$$

$$= \int_0^H |\bigcup_i E_i(t)|\, dt \ge A(\mathcal{Q}) \int_0^H \Psi(\rho(t))\, dt. \tag{28}$$

Normalized Jensen on $[0, H]$ and a second layer-cake computation yield

$$\int_0^H \Psi(\rho(t))\, dt \ge H\Psi\left(\frac{1}{H}\int_0^H \rho(t)\, dt\right),$$

$$\int_0^H \rho(t)\, dt = \frac{1}{D}\sum_i \int_{\mathbb{R}^d} f_i(z)\, dz = \frac{1}{D}\sum_i |S_i|.$$

Substitution in (28) proves (25). The normalization uses the fixed body mass HD rather than the shaded mass. Consequently the same argument includes the zero-density case, where $\Psi(0) = 0$. □

Repeated indices pass through the lift unchanged: the union and fibre maximum are geometric, whereas D and the density numerator retain the indexed sums. The lower-dimensional estimate must therefore already be valid for the indexed family. Section 4 establishes precisely that input before the lift is applied.

Proposition 3.2 (Product sharpness)**.** *Fix measurable $E_i \subset Q_i$ and take $S_i = I \times E_i$. Then every inequality in the fibre and Jensen part of the proof of Theorem 3.1 is an equality. Consequently, neither the factor H nor the response at the mean density can be improved uniformly by the lifting argument alone.*

Proof. For almost every $0 < t < H$, one has $f_i = H\mathbf{1}_{E_i}$ and $E_i(t) = E_i$. Therefore

$$|\bigcup_i S_i| = H|\bigcup_i E_i|, \qquad \rho(t) = \frac{\sum_i |E_i|}{D}.$$

The fibre comparison and Jensen are both equalities. Any uniform improvement in (25) would therefore give the same improvement in (24). □

Remark 3.3 (Nonconvex responses). Without convexity, (28) remains valid, but the value of Ψ at the mean need not be controlled. If $\mathrm{co}(\Psi)$ denotes the lower convex envelope, the automatic conclusion is

$$|\bigcup_i S_i| \geq HA(\mathcal{Q})\ \mathrm{co}(\Psi)\left(\frac{\sum_i |S_i|}{HD}\right).$$

In the application below, $\Psi(t) = t^{K_\varepsilon}$ with $K_\varepsilon \geq 1$, so the response is convex and its exponent is preserved exactly.

4. Indexed convex unions in three dimensions

Projection creates the first indexing difficulty in the local argument. The three-dimensional convex-union theorem is formulated for essentially-distinct bodies, while two different four-dimensional child indices may have the same projection. Passing to a geometric set of representatives would discard precisely the multiplicities that the final factorization must retain. We instead derive the localized essentially-distinct estimate needed here, organize coincident projections by affine-adapted canonical boxes, and recover the indexed normalization through a collision-layer cancellation.

4.1. **The essentially-distinct input.** Two measurable sets $Q, Q' \subset \mathbb{R}^3$ are *essentially distinct* if

$$|Q \cap Q'| \leq \frac{1}{2}\max(|Q|, |Q'|). \tag{29}$$

This is the convention of Wang and Zahl [10]. We begin with their set-valued hypothesis; the indexed extension is proved later in this section.

Proposition 4.1 (Localized essentially-distinct union estimate)**.** *Let $0 < q_1 \leq q_2 \leq q_3$, and let $\mathcal{Q}$ be a nonempty essentially-distinct family of convex bodies contained in a ball of radius $O(q_3)$. Suppose their ordered John semiaxes are comparable, with fixed constants, to q_1, q_2, q_3. For measurable $E(Q) \subset Q$, put*

$$\rho = \frac{\sum_{Q\in\mathcal{Q}} |E(Q)|}{\sum_{Q\in\mathcal{Q}} |Q|}, \qquad m = \Delta_{\max}(\mathcal{Q}).$$

For every $\varepsilon > 0$ there is $K_\varepsilon \geq 1$ such that

$$|\bigcup_{Q\in\mathcal{Q}} E(Q)| \gtrsim_\varepsilon \left(\frac{q_1}{q_3}\right)^\varepsilon \rho^{K_\varepsilon} m^{-1} \sum_{Q\in\mathcal{Q}} |Q|. \tag{30}$$

The constants depend on the fixed axis-comparability constants and on the fixed constant hidden in the ambient radius $O(q_3)$.

Proof. It suffices to prove the claim for $0 < \varepsilon \le 1$, since for $0 < a \le 1$ the bound with exponent $\min\{\varepsilon, 1\}$ implies the bound with exponent ε. After an affine scaling of all lengths by q_3^{-1}, write $a = q_1/q_3$, $b = q_2/q_3$, and let $N = \#\mathcal{Q}$ and $v \asymp ab$ be the common volume. We use the positive-parameter assertion $F(t,t)$ for $0 < t \le 2/3$. In the source notation, it is obtained as follows. Their Theorem 1.9 gives $D(0,0)$; at $\sigma = \omega = t$, Definition 1.5 makes $D(t,t)$ weaker by the factor $(\#\mathcal{T})^{-t} \le 1$. Proposition 1.6 gives $E(t,t)$, and Proposition 5.14 gives $F(t,t)$ [10]. In the source notation it gives, for families of density at least a^η, where $\eta := \eta_{\mathrm{WZ}}(\varepsilon/2; t, t) > 0$,

$$|\bigcup_{Q} E(Q)| \gtrsim a^{\varepsilon/2} b^t m^{-1} N v (m^{-3/2} \ell N v^{1/2})^{-t} \mathcal{D}(\mathcal{Q})^{-t}, \tag{31}$$

where ℓ is the source's Frostman slab constant and $\mathcal{D}$ is its geometric concentration quantity.

The localization gives $Nv \lesssim m$, while testing m on a member of the family gives $m \gtrsim 1$. A slab containing an $a \times b \times 1$ John-normalized body has thickness and volume $\gtrsim a$, and hence $\ell \lesssim a^{-1}$. For every Q, the localized subfamily $\mathcal{Q}[N_b(Q)]$ has Katz–Tao constant at most m by subfamily monotonicity. Remark 5.6, Situation 2, therefore gives $\mathcal{D}(\mathcal{Q}) \lesssim m^{1/2}$. Consequently, with $B := m^{-3/2} \ell N v^{1/2}$,

$$B\mathcal{D}(\mathcal{Q}) \lesssim m^{-1} \ell N v^{1/2} \lesssim \ell v^{-1/2} \lesssim a^{-2}.$$

Choose $t = \varepsilon/10$. Since $b \ge a$,

$$b^t B^{-t} \mathcal{D}(\mathcal{Q})^{-t} = b^t \big(B\mathcal{D}(\mathcal{Q})\big)^{-t} \gtrsim b^t a^{2t} \ge a^{3t}.$$

For $0 < a \le 1$, the exponent $\varepsilon/2 + 3\varepsilon/10 = 4\varepsilon/5$ is smaller than ε; thus (31) is at least the right side of (30) whenever $\rho \ge a^\eta$, because $\rho^K \le 1$.

It remains to remove the source density threshold. If $0 < \rho < a^\eta$, one shading has measure $\gtrsim \rho v$, and $v \asymp ab \ge a^2$. Since $m^{-1} N v \lesssim 1$, any

$$K_\varepsilon \ge 1 + \frac{2}{\eta}$$

satisfies

$$|\bigcup_{Q} E(Q)| \gtrsim \rho a^2 \gtrsim a^\varepsilon \rho^{K_\varepsilon} m^{-1} N v.$$

The zero-density case is immediate, completing the all-density localized estimate. The source's "outer John ellipsoid" is used with the fixed-dilate convention of Section 2, which changes only the fixed comparability constants. □

Thus Proposition 4.1 is the localized all-density consequence needed below. Its reduction keeps the positive source parameters explicit and supplies the remaining small-density range by the largest-shading argument. Appendix B records the full notation dictionary for this derivation.

4.2. **Affine-adapted canonical boxes.** A Euclidean net in all Euler angles does not give aspect-ratio-uniform collision control. For example, boxes of shape $\delta \times \delta \times 1$ with a common center and long axis, rotated in roll increments of size δ, form a clique of size comparable to δ^{-1} in the non-essential-distinctness graph. The parameter metric must therefore measure the relative affine transform at the scale of the box itself.

We first isolate the parameter-space fact used in the discretization. Put $C = [-1, 1]^3$ and, for a fixed positive diagonal matrix D, write

$$B(c, U) = c + UDC, \qquad c \in \mathbb{R}^3, \quad U \in SO(3).$$

Let

$$H_D = \{S \in SO(3) : SDC = DC\}.$$

This is the orientation-preserving signed-permutation symmetry group of the rectangular box, and $\#H_D \le 24$. For two frame representations define

$$A = D^{-1}U^T U' D, \qquad b = D^{-1}U^T(c' - c),$$

Here $|\cdot|$ is the Euclidean norm on vectors and $\|\cdot\|$ is its induced operator norm on matrices. Both norms are fixed before D is chosen, so all dependence on the affine shape appears explicitly through the conjugation by D. Set

$$d_0((c, U), (c', U')) = \max\{\|A - I\|, \|A^{-1} - I\|, |b|, |A^{-1}b|\}. \tag{32}$$

Minimize (32) over the left and right actions of H_D and denote the resulting symmetric gauge on geometric boxes by d_D.

Lemma 4.2 (Local affine parameter packing)**.** *There are dimensional constants $\eta_* > 0$ and $C < \infty$, independent of the condition number of D, with the following properties.*

(i) *The gauge d_D is independent of the frame representations of the two geometric boxes. If $d_D(B, B') \le \eta \le \eta_*$, then, after choosing representatives realizing the minimum,*

$$B' \subset (1 + C\eta)B, \qquad B \subset (1 + C\eta)B',$$

where the dilations are about the respective centers.

(ii) *If LB denotes dilation about the center by $L \ge 1$, then*

$$d_{LD}(LB, LB') \ge L^{-1} d_D(B, B').$$

(iii) *Fix B_0 of shape D and $0 < \zeta \le 1$. If boxes B_β of the same shape satisfy $d_D(B_\beta, B_\gamma) > \zeta$ for $\beta \ne \gamma$ and*

$$|B_0 \cap B_\beta| > \tfrac{1}{2}|B_0|,$$

then their number is $O(\zeta^{-12})$, with a dimensional implicit constant.

Proof. All frame representations of one chosen geometric box differ by right multiplication by an element of H_D. Such an element is a signed permutation among equal diagonal entries of D, so it commutes with D. Consequently a change of the two representatives sends (b, A) to one of the declared finite left/right transforms. The minimum is therefore a function of the geometric

boxes. Swapping the boxes sends (b, A) to $(-A^{-1}b, A^{-1})$, so the four terms in (32) also prove symmetry.

In the coordinates of the first box, the second is $b + AC$. If the gauge is at most η, choose minimizing representatives. For $x \in C$,

$$b + Ax = x + \big(b + (A - I)x\big) \in (1 + C\eta)C.$$

The two inverse terms give the reverse containment, establishing (i). Under common dilation, A is unchanged and b is replaced by b/L. For each fixed pair of symmetry representatives, the maximum in (32) decreases by at most the factor L; taking the finite minimum proves (ii).

For (iii), normalize B_0 affinely to C. A neighbor has the form $b + AC$ with $\det A = 1$. The set

$$\Omega = \{y \in C : b + Ay \in C\}$$

has measure greater than $|C|/2$. If a_r is a row of A, then Ω lies in the slab $\{y \in C : |b_r + a_r \cdot y| \le 1\}$. Slicing in a coordinate on which a_r has largest absolute component shows that the volume of this slab inside C is $O(1/\|a_r\|_\infty)$. Thus every row of A is bounded. Nonempty intersection gives $b = x - Ay$ for some $x, y \in C$, and hence bounds b. Since $\det A = 1$, the cofactor formula bounds A^{-1} as well. All neighboring parameters (b, A) therefore lie in one fixed compact subset of $\mathbb{R}^{12}$.

Choose for each neighbor a representative in this central chart. For two such parameters,

$$A_{\beta\gamma} = A_\beta^{-1} A_\gamma, \qquad b_{\beta\gamma} = A_\beta^{-1}(b_\gamma - b_\beta).$$

On the fixed compact set, the identity-representative value of d_0 is at most a dimensional constant times $|b_\beta - b_\gamma| + \|A_\beta - A_\gamma\|$. Since d_D is the minimum over representatives, $d_D(B_\beta, B_\gamma) > \zeta$ forces ordinary Euclidean separation $\gtrsim \zeta$ of these parameters. Euclidean packing in the compact subset of $\mathbb{R}^{12}$ now gives $O(\zeta^{-12})$ neighbors. □

Corollary 4.3 (Condition-number-independent collision degree)**.** *Let D be any positive diagonal matrix and let $0 < \zeta \le 1$. If a family of shape-D boxes is ζ-separated in d_D, then the graph joining two boxes whose intersection has more than half the volume of either box has maximum degree $O(\zeta^{-12})$. The implicit constant is dimensional and is independent of $\|D\|\|D^{-1}\|$.*

Proof. Fix a vertex B_0. All boxes have the same volume, so every neighbor satisfies the intersection hypothesis in Lemma 4.2(iii). That lemma counts the neighbors of B_0 by $O(\zeta^{-12})$, uniformly in D. □

Lemma 4.4 (Canonical boxes)**.** *Fix $C_0 \ge 1$. Let $(Q_i)_{i \in I}$ be a finite indexed family of convex bodies in $\mathbb{R}^3$ whose ordered John semiaxes lie in $[q_j, C_0 q_j]$, $1 \le j \le 3$. There are an assignment $i \mapsto \alpha(i)$ and boxes R_α such that:*

(i) *$Q_i \subset R_{\alpha(i)}$ and $|Q_i| \asymp_{C_0} |R_{\alpha(i)}|$;*

(ii) *the distinct geometric boxes R_α can be colored with $O_{C_0}(1)$ colors so that boxes of one color are essentially distinct;*

(iii) *the number of colors is independent of* q_3/q_1.

Proof. Fix a small dimensional $\eta_0 > 0$ and round each John semiaxis upward to the least number of the form $q_j(1+\eta_0)^k$. There are at most

$$\left(1 + \left\lceil \log_{1+\eta_0} C_0 \right\rceil\right)^3 \tag{33}$$

rounded triples. We use disjoint color palettes for different triples and work with one exact triple. Choose an arbitrary John eigenframe for each member; when eigenvalues repeat this is merely a tie-break, and all estimates below are uniform in that choice. After incorporating a fixed John dilation, every member lies in a box

$$B(c,U) = c + UD[-1,1]^3, \qquad D = \operatorname{diag}(q_1,q_2,q_3),$$

of comparable volume.

The group H_D in Lemma 4.2 identifies only different frame representations of a resulting geometric box; it is not being used to remove the continuous eigenframe tie-break. Choose greedily a maximal η-separated set of distinct input boxes in the symmetric gauge d_D, where $\eta > 0$ is a sufficiently small dimensional constant. Maximality assigns every input box B_i to a selected box B^0_α at distance at most η. Lemma 4.2(i) gives

$$B_i \subset LB^0_\alpha, \qquad L = 1 + C\eta.$$

Set $R_\alpha = LB^0_\alpha$. The containment and volume comparison follow. Common dilation leaves the selected boxes at least $L^{-1}\eta$-separated by Lemma 4.2(ii). If R_β is not essentially distinct from R_α, their equal volumes give $|R_\alpha \cap R_\beta| > |R_\alpha|/2$. Corollary 4.3 bounds the degree of this collision graph independently of the aspect ratio. A bounded greedy coloring, followed by the finite number of palettes in (33), completes the proof. □

4.3. **Collision cancellation.** We now return to the original indices. Canonical boxes merge geometrically close projections, but the size of a collision cluster is retained as indexed mass. On a layer where the cluster sizes are comparable, the same cluster size improves the Katz–Tao density of the geometric representatives. These two effects cancel in the union estimate.

For an indexed family $\mathcal{Q} = (Q_i)_{i\in I}$ with comparable John dimensions, choose canonical boxes as in Lemma 4.4. Let

$$I_\alpha = \{i : \alpha(i) = \alpha\}, \quad n_\alpha = \#I_\alpha, \quad F_\alpha = \bigcup_{i\in I_\alpha} E_i,$$

and put

$$S = \sum_i |Q_i|, \qquad M = \sum_i |E_i|, \qquad \rho = M/S, \qquad m = \Delta_{\max}(\mathcal{Q}).$$

Lemma 4.5 (Density gain on a collision layer)**.** *If* $n \le n_\alpha < 2n$ *on a set* $\mathcal{A}_n$ *of canonical boxes, then*

$$\Delta_{\max}(\{R_\alpha : \alpha \in \mathcal{A}_n\}) \lesssim m/n. \tag{34}$$

Proof. For every convex body K, containment and two-sided volume comparison give

$$\sum_{\substack{\alpha\in\mathcal{A}_n\\ R_\alpha\subset K}} |R_\alpha| \lesssim \frac{1}{n} \sum_{\substack{\alpha\in\mathcal{A}_n\\ R_\alpha\subset K}} \sum_{i\in I_\alpha} |Q_i| \le \frac{1}{n} \sum_{i:Q_i\subset K} |Q_i| \le \frac{m}{n}|K|.$$

The middle implication uses only $R_\alpha \subset K \Rightarrow Q_i \subset R_\alpha \subset K$. □

Theorem 4.6 (Essentially-distinct to indexed)**.** *Under the hypotheses of Proposition 4.1, but allowing arbitrary indexed repetitions, let $N = \#I$ and $L_N = 1 + \lceil \log_2 N \rceil$. Then*

$$|\bigcup_i E_i| \gtrsim_\varepsilon \left(\frac{q_1}{q_3}\right)^\varepsilon L_N^{-1} \rho^{K_\varepsilon} m^{-1} \sum_i |Q_i|. \tag{35}$$

The estimate remains true without a common ambient ball.

Proof. First suppose that the family is localized. On a collision layer $\mathcal{A}_n = \{\alpha : n \le n_\alpha < 2n\}$, write S_n, M_n for the body and shading masses of the corresponding indices and $\rho_n = M_n/S_n$. The elementary estimate

$$|F_\alpha| \ge n_\alpha^{-1} \sum_{i\in I_\alpha} |E_i|$$

and volume comparison imply

$$\sum_{\alpha\in\mathcal{A}_n} |R_\alpha| \asymp S_n/n, \qquad \sum_{\alpha\in\mathcal{A}_n} |F_\alpha| \gtrsim M_n/n. \tag{36}$$

Split the canonical boxes into the bounded number of colors supplied by Lemma 4.4. For a color ν, let $S_{n,\nu}$ and $M_{n,\nu}$ be the body and shading masses of its assigned input indices, and put $\rho_{n,\nu} = M_{n,\nu}/S_{n,\nu}$ when $S_{n,\nu} > 0$. Within that color, Proposition 4.1, Lemma 4.5, and (36) give the lower bound

$$\left(\frac{q_1}{q_3}\right)^\varepsilon m^{-1} S_{n,\nu} \rho_{n,\nu}^{K_\varepsilon}$$

for the union of its merged shadings. The color unions need not be disjoint, so we do not add these bounds. Instead, weighted Jensen gives

$$\sum_\nu S_{n,\nu} \rho_{n,\nu}^{K_\varepsilon} \ge S_n \rho_n^{K_\varepsilon}.$$

Because the number of colors is bounded, one color ν_* contributes a fixed fraction of the right-hand side. The union over the entire collision layer contains the union for ν_*, and therefore

$$|\bigcup_{\alpha\in\mathcal{A}_n} F_\alpha| \gtrsim_\varepsilon \left(\frac{q_1}{q_3}\right)^\varepsilon \rho_n^{K_\varepsilon} m^{-1} S_n. \tag{37}$$

At the level of powers, the factor $1/n$ in canonical-box mass is canceled by the factor n in $(m/n)^{-1}$.

There are at most L_N nonempty layers. Since $K_\varepsilon \geq 1$, weighted Jensen gives

$$\sum_n S_n \rho_n^{K_\varepsilon} \geq S \left(\sum_n \frac{S_n}{S} \rho_n \right)^{K_\varepsilon} = S \rho^{K_\varepsilon}. \tag{38}$$

Hence one layer contributes at least $L_N^{-1} S \rho^{K_\varepsilon}$, and (35) follows from (37).

For a nonlocalized family, assign each body by its John center to a cube of a q_3-lattice. A fixed enlargement of that cube contains the whole body. Color the lattice with a dimensional number of colors so that equal-color enlargements are disjoint. Apply the localized argument in each cube. Within a fixed lattice color, the resulting union lower bounds add because the enlargements are disjoint; weighted Jensen combines their body-mass densities. Across lattice colors the unions can overlap, so weighted Jensen followed by a bounded-color pigeonhole selects one color; the full union contains that selected color's union. This costs only a dimensional constant. The global bound m dominates each local Katz–Tao density, which gives the final assertion. □

Complete duplication illustrates the normalization. If all $Q_i = Q$ and all $E_i = E$, then $m = N$, and the right side of (35) is bounded by a constant multiple of $\rho^{K_\varepsilon}|Q| \leq |E|$. If the shadings are disjoint, the cluster union bound is conservative but valid. Throughout the construction the original indices remain in the mass and Katz–Tao quantities.

4.4. **The local Frostman form.** The preceding theorem assumes comparable John dimensions. For the projected family arising in Section 5, each body instead contains an h-ball and lies in a common box of longest side R. Dyadic decomposition of the three John axes reduces this more flexible setting to Theorem 4.6; relative Frostman control supplies the uniform Katz–Tao bound in every bin.

Theorem 4.7 (Indexed local Frostman union estimate)**.** *Let $B' \subset \mathbb{R}^3$ be a box of longest side R, and let $\mathcal{Q} = (Q_i)_{i \in I}$ be a nonempty finite indexed family of convex bodies contained in B'. Suppose every Q_i contains a ball of radius h, has diameter $O(R)$, and $C_F(\mathcal{Q}, B') \leq C_F$. For measurable $E_i \subset Q_i$, let*

$$\rho = \frac{\sum_i |E_i|}{\sum_i |Q_i|}, \qquad N = \#I,$$

and set

$$L_N = 1 + \lceil \log_2 N \rceil, \qquad J_h = \left(1 + \lceil \log_+(R/h) \rceil\right)^3.$$

Then, for every $\varepsilon > 0$,

$$\Big|\bigcup_i E_i\Big| \gtrsim_\varepsilon \left(\frac{h}{R}\right)^\varepsilon L_N^{-1} J_h^{-1} C_F^{-1} \rho^{K_\varepsilon} |B'|. \tag{39}$$

Proof. Put $S = \sum_i |Q_i|$ and $D = S/|B'|$. The relative Frostman hypothesis implies the absolute indexed bound

$$\Delta_{\max}(\mathcal{Q}_0) \le C_F D \tag{40}$$

for every subfamily $\mathcal{Q}_0$. Indeed, for a convex test body K, the bodies counted in K are also counted in $K \cap B'$, while $|K| \ge |K \cap B'|$; use (2) and subfamily monotonicity.

Because Q_i contains an h-ball and lies in B', its shortest John semiaxis is $\gtrsim h$ and its longest semiaxis is $O(R)$. Divide the three ordered semiaxes into dyadic triples. There are at most $O(J_h)$ nonempty bins. For bin j, let S_j, M_j be its body and shading masses and $\rho_j = M_j/S_j$. Theorem 4.6 and (40) give

$$\Big|\bigcup_{i \text{ in bin } j} E_i\Big| \gtrsim_\varepsilon \left(\frac{h}{R}\right)^{\varepsilon} L_N^{-1}(C_F D)^{-1} S_j \rho_j^{K_\varepsilon} = \left(\frac{h}{R}\right)^{\varepsilon} L_N^{-1} C_F^{-1} |B'| \frac{S_j}{S} \rho_j^{K_\varepsilon}.$$

Finally,

$$\sum_j \frac{S_j}{S} \rho_j^{K_\varepsilon} \ge \left(\sum_j \frac{S_j}{S} \rho_j\right)^{K_\varepsilon} = \rho^{K_\varepsilon}.$$

One of the at most $O(J_h)$ bins therefore gives (39). When $h \gtrsim R$ there is one bin, and the same proof applies. Empty shadings give zero right side. □

The denominator in Theorem 4.7 is the body mass of the fixed indexed family, including indices whose current shading is empty. Consequently the same estimate can be applied to every measurable refinement produced later by the cellular construction.

5. The local four-dimensional transfer

We now convert the indexed three-dimensional estimate into a parentwise statement in four dimensions. The main issue is that thickening children of different geometry produces different volume weights. We first quantify thickening for arbitrary measurable shadings, then project along a shortest parent direction so that the volume of each thickened body is comparable to h times its projected volume. The resulting weighted Frostman data can be passed to Theorem 4.7 and lifted back by Theorem 3.1.

5.1. Convex thickening.

The cellular argument will apply the local estimate after several measurable refinements. We therefore need a thickening bound that depends on the density of an arbitrary measurable shading, rather than on any regularity of that shading.

Lemma 5.1 (Thickening an arbitrary measurable shading)**.** *For every convex body $V \subset \mathbb{R}^n$, every measurable $Y \subset V$, and every $h > 0$,*

$$|N_h(Y)| \ge c_n \frac{|Y|}{|V|} |N_h(V)|. \tag{41}$$

Proof. Use John coordinates with semiaxes $a_1, \dots, a_n$. From (1) and a containing rectangular box,

$$|V| \asymp_n \prod_j a_j, \qquad |N_h(V)| \lesssim_n \prod_j (a_j + h). \tag{42}$$

Let L be the span of the axes with $a_j \geq h$ and $S = L^\perp$. Write $m = \dim S$ and let Y_x be the S-fibre over $x \in L$. The John box bounds every fibre by $C_n \prod_{a_j<h} a_j$. If $X = \{x : |Y_x| > 0\}$, Fubini gives

$$|X| \gtrsim_n \frac{|Y|}{\prod_{a_j<h} a_j}. \tag{43}$$

For every $x \in X$, the corresponding fibre of $N_h(Y)$ contains the h-neighborhood in S of a nonempty set and hence has m-volume at least $\omega_m h^m$. Integrating,

$$|N_h(Y)| \gtrsim_n |Y| \prod_{a_j<h} \frac{h}{a_j}. \tag{44}$$

When $m = 0$, this is simply $|N_h(Y)| \geq |Y|$. On the other hand, (42) gives

$$\frac{|Y|}{|V|} |N_h(V)| \lesssim_n |Y| \prod_{a_j<h} \frac{h}{a_j}.$$

Together with (44), this proves the result at every intermediate scale. If Y is null, the right side of (41) is zero. □

The same John-box argument, now using a box inside E_V and a box inside the Euclidean h-ball, gives the two-sided estimate

$$|N_h(V)| \asymp_n \prod_{j=1}^{n} (a_j(V) + h). \tag{45}$$

Lemma 5.2 (Buffered parent stability)**.** *Let (W_α) be a finite indexed family in $\mathbb{R}^n$, let $0 < r_\alpha \leq C_0 a_1(W_\alpha)$, and put $W_\alpha^\sharp = N_{r_\alpha}(W_\alpha)$. Then*

$$|W_\alpha^\sharp| \asymp_{n,C_0} |W_\alpha|, \qquad \Delta_{\max}(\{W_\alpha^\sharp\}) \lesssim_{n,C_0} \Delta_{\max}(\{W_\alpha\}). \tag{46}$$

Proof. The volume comparison follows from (45). If $W_\alpha^\sharp \subset K$, then $W_\alpha \subset K$, and hence

$$\sum_{\alpha : W_\alpha^\sharp \subset K} |W_\alpha^\sharp| \lesssim \sum_{\alpha : W_\alpha \subset K} |W_\alpha| \leq \Delta_{\max}(\{W_\alpha\}) |K|.$$

Take the supremum over positive-volume convex K. Every sum retains the parent indices. □

The buffer lemma allows the coarse shading to live in a genuine neighborhood of the parent while preserving both its volume scale and its indexed Katz–Tao density. Section 8 explains why an intersection with the original parent would not provide an aspect-uniform substitute.

5.2. **Shortest-direction projection.** Projection is useful only if the lost fibre direction has length comparable to the analytic radius. Choosing a shortest John direction of the parent provides this comparison simultaneously for every child, regardless of the child's own orientation.

Lemma 5.3 (Projection–volume comparison)**.** *Let e be a unit vector, let π be orthogonal projection onto $e^{\perp}$, and set*

$$P = N_h(V), \qquad Q = \pi(P).$$

Then

$$c_n h|Q| \le |P| \le (\mathrm{width}_e(V) + 2h)|Q|. \tag{47}$$

In particular, if $V \subset W$ and $\mathrm{width}_e(W) \le C_0 h$, then $|P| \asymp_{n,C_0} h|Q|$.

Proof. Let $K = \pi(V)$. Orthogonal projection commutes with Minkowski addition by a ball, so $Q = K + B_{e^{\perp}}(0,h)$. If $z \in K + B(0,h/2)$, choose $x \in K$ with $|z-x| < h/2$ and a point $(t,x) \in V$ in coordinates $\mathbb{R}e \times e^{\perp}$. The fibre over z of the h-ball around (t,x) has length at least $\sqrt{3}h$. Thus

$$|P| \ge \sqrt{3}h|K + B(0,h/2)|.$$

Convexity, with any $x_0 \in K$, gives $K + B(0,h) \subset x_0 + 2[(K + B(0,h/2)) - x_0]$; hence the two projected volumes differ by at most 2^{n-1}, giving the lower bound. The entire e-coordinate range of P has length at most $\mathrm{width}_e(V) + 2h$, which proves the upper bound by Fubini. □

Let $W \subset \mathbb{R}^4$ have John semiaxes $u_1 \le u_2 \le u_3 \le u_4$, choose a shortest John direction e_1, and let $h = c_0 u_1$. Since $W \subset 4E_W$, $\mathrm{width}_{e_1}(W) \le 8u_1 = 8h/c_0$. Therefore Lemma 5.3 applies uniformly to every $V \subset W$, without any condition on the orientation of V. Moreover, in the same coordinates the product box

$$B = I \times B' = \prod_{j=1}^{4} [c_j - (4u_j + h), c_j + (4u_j + h)] \tag{48}$$

contains $N_h(W)$ and satisfies

$$H := |I| \asymp_{c_0} h, \qquad R := \mathrm{longside}(B') \asymp_{c_0} u_4, \qquad |B| \asymp_{c_0} |W|. \tag{49}$$

5.3. **Mean-normalized weighted Frostman inheritance.** Thickening replaces the body mass $|V|$ by $r_W(V)|V|$. The relative Frostman coefficient must therefore be computed with the same weights. The next two lemmas compare this weighted coefficient with the original family and show that it survives shortest-direction projection.

Lemma 5.4 (Comparison with the unweighted coefficient)**.** *Let $\mathcal{V}_W$ be nonempty and $C_F(\mathcal{V}_W, W) \le C$. Then*

$$C^{\mathrm{wt}}_{F,h}(\mathcal{V}_W, W) \le C \frac{r^{+}_W}{\overline{r}_W}. \tag{50}$$

Proof. For a convex $K \subset W$,

$$\sum_{V \subset K} r_W(V)|V| \le r_W^+ \sum_{V \subset K} |V| \le r_W^+ C \frac{T_W}{|W|}|K|.$$

The weighted total density equals

$$\frac{\sum_V r_W(V)|V|}{|W|} = \overline{r}_W \frac{T_W}{|W|}.$$

Division proves (50). This concentration step uses the maximum-to-mean ratio; the minimum inflation enters separately when shaded density is converted in Corollary 5.7. □

Lemma 5.5 (Weighted thickening and projection inheritance)**.** *Let (V_i) be a nonempty indexed family in a convex body W, put $P_i = N_h(V_i) \subset B$, assume $|B| \le C_B|W|$, and write $r_i = |P_i|/|V_i|$, $r_+ = \max_i r_i$, and $\overline{r} = (\sum_i r_i|V_i|)/(\sum_i |V_i|)$. Then*

$$C_F(\{P_i\}, B) \le C_B C_{F,h}^{\mathrm{wt}}(\{V_i\}, W). \tag{51}$$

If $C_F(\{V_i\}, W) \le C$, then

$$C_F(\{P_i\}, B) \le C_B C \frac{r_+}{\overline{r}}. \tag{52}$$

Suppose also that $B = I \times B'$, $Q_i = \pi(P_i) \subset B'$, and

$$c_P h|Q_i| \le |P_i| \le C_P h|Q_i|.$$

Then

$$C_F(\{Q_i\}, B') \le \frac{C_P}{c_P} C_F(\{P_i\}, B). \tag{53}$$

Proof. Set

$$D_r = \frac{\sum_i r_i|V_i|}{|W|}, \qquad D_P = \frac{\sum_i |P_i|}{|B|}.$$

Then $D_P = D_r|W|/|B| \ge C_B^{-1} D_r$. If $P_i \subset K \subset B$, then $V_i \subset K \cap W$. If the relevant sum is nonzero, $K \cap W$ is a positive-volume convex body, and Definition 2.1 gives

$$\sum_{P_i \subset K} |P_i| = \sum_{P_i \subset K} r_i|V_i| \le C_{F,h}^{\mathrm{wt}} D_r |K \cap W| \le C_{F,h}^{\mathrm{wt}} D_r |K|.$$

The zero-sum case is immediate. Division by $D_P|K|$ proves (51); Lemma 5.4 gives (52).

For projection, $Q_i \subset K' \subset B'$ implies $P_i \subset I \times K'$. With $H = |I|$,

$$\sum_{Q_i \subset K'} |Q_i| \le \frac{C_F(\{P_i\}, B) D_P H |K'|}{c_P h}, \qquad D_Q := \frac{\sum_i |Q_i|}{|B'|} \ge \frac{D_P H}{C_P h}.$$

After division, the factors H/h cancel exactly, proving (53). □

5.4. **The nonuniform local theorem.** We can now keep the full inflation profile in the local response. The theorem below is stated for a fixed base family so that its weighted coefficient and denominators remain available uniformly after any later refinement.

Theorem 5.6 (Intrinsic nonuniform local transfer)**.** *Let $W \subset \mathbb{R}^4$ have John semiaxes $u_1 \le u_2 \le u_3 \le u_4$, fix $c_0 > 0$, and put $h = c_0 u_1$. Let $\mathcal{V}_W = (V_i)_{i\in J}$ be an arbitrary nonempty finite indexed family of convex bodies $V_i \subset W$. Write $r_i = |N_h(V_i)|/|V_i|$ and $\overline{r} = (\sum_i r_i|V_i|)/(\sum_i |V_i|)$. For measurable $Y_i \subset V_i$, use (4) and Definition 2.1, and put*

$$\delta_W = \frac{u_1}{u_4}, \qquad \mathfrak{L}_W = 2 + \log(2 + \#J) + \log(2 + u_4/u_1).$$

For every $\varepsilon > 0$ there are $K_\varepsilon \ge 2$ and $A_{0,\varepsilon} < \infty$ such that

$$|N_h(\bigcup_i Y_i)| \gtrsim_{\varepsilon,c_0} \mathfrak{L}_W^{-A_{0,\varepsilon}} \delta_W^{\varepsilon} \big(C^{\mathrm{wt}}_{F,h}(\mathcal{V}_W, W)\big)^{-1} \big(d^{\mathrm{wt}}_{W,h}(Y)\big)^{K_\varepsilon} |W|. \tag{54}$$

The estimate is uniform over every measurable refinement of the fixed base family; the quantities r_i, $\overline{r}$, $C^{\mathrm{wt}}_{F,h}$, and T_W remain frozen.

Proof. Set $P_i = N_h(V_i)$ and $S_i = N_h(Y_i)$. For a finite family,

$$\bigcup_i S_i = N_h(\bigcup_i Y_i). \tag{55}$$

Lemma 5.1 gives

$$\sum_i |S_i| \gtrsim \sum_i r_i |Y_i|.$$

Use the shortest-direction product box (48), let π be projection onto $e_1^{\perp}$, and put $Q_i = \pi(P_i)$. Then $S_i \subset P_i \subset I \times Q_i$ and

$$|P_i| \asymp_{c_0} h|Q_i|. \tag{56}$$

Equations (51) and (53) give

$$C_F(\{Q_i\}, B') \lesssim_{c_0} C^{\mathrm{wt}}_{F,h}(\mathcal{V}_W, W). \tag{57}$$

Each Q_i contains a three-ball of radius h and has diameter $O(R)$.

Apply Theorem 4.7 to arbitrary measurable $E_i \subset Q_i$. If its exponent is less than 2, replace it by 2: since its density lies in $[0,1]$, this only weakens the lower bound. The explicit cardinality and projected-axis losses are bounded below by $\mathfrak{L}_W^{-A_{0,\varepsilon}}$. Theorem 3.1, with $\Psi(t) = t^{K_\varepsilon}$, then gives

$$|\bigcup_i S_i| \gtrsim_{\varepsilon,c_0} H|B'|\, \mathfrak{L}_W^{-A_{0,\varepsilon}} \left(\frac{h}{R}\right)^{\varepsilon} (C^{\mathrm{wt}}_{F,h})^{-1} \left(\frac{\sum_i |S_i|}{H \sum_i |Q_i|}\right)^{K_\varepsilon}. \tag{58}$$

By (56) and $H \asymp h$,

$$\frac{\sum_i |S_i|}{H \sum_i |Q_i|} \gtrsim \frac{\sum_i r_i |Y_i|}{\sum_i r_i |V_i|} = d^{\mathrm{wt}}_{W,h}(Y).$$

Finally, $H|B'| \asymp |W|$ and $h/R \asymp u_1/u_4$ by (49); use (55). Since every Q_i contains an h-ball and lies in a box of longest side $O(u_4)$, the dyadic axis

ranges are controlled by u_4/h. This gives the stated local logarithm without introducing a child minimum scale. □

Corollary 5.7 (Unweighted local efficiencies)**.** *In the setting of Theorem 5.6,*

$$|N_h(\bigcup_i Y_i)| \gtrsim_{\varepsilon,c_0} \mathfrak{L}_W^{-A_{0,\varepsilon}} \delta_W^{\varepsilon} \kappa_W^{(K_\varepsilon)} d_W(Y)^{K_\varepsilon} |W|. \tag{59}$$

If $C_F(\mathcal{V}_W, W) \le C$*, then*

$$|N_h(\bigcup_i Y_i)| \gtrsim_{\varepsilon,c_0} \mathfrak{L}_W^{-A_{0,\varepsilon}} \delta_W^{\varepsilon} C^{-1} \eta_W^{(K_\varepsilon)} d_W(Y)^{K_\varepsilon} |W|. \tag{60}$$

Proof. Insert (5) into (54) to obtain (59). Lemma 5.4 gives

$$(C_{F,h}^{\mathrm{wt}})^{-1} \left(\frac{r_W^-}{\overline{r}_W}\right)^{K_\varepsilon} \ge C^{-1} \frac{\overline{r}_W}{r_W^+} \left(\frac{r_W^-}{\overline{r}_W}\right)^{K_\varepsilon},$$

which is (60). □

The two factors in the elementary coefficient have different origins. Weighted Frostman inheritance contributes $\overline{r}_W/r_W^+$ once, whereas conversion from the original shaded density to the weighted density contributes $r_W^-/\overline{r}_W$ with the response exponent K_ε. Keeping these mechanisms separate yields the mean-normalized expression in (60).

6. Weighted profile assembly and cellular factoring

The local theorem supplies a response for each parent, but the child loads D_W and the local efficiencies α_W need not be comparable. Selecting a common load level would change the coarse normalization of the main theorem. The weighted Hölder inequality below instead measures the cost of assembling all parents in a single explicit coefficient. The cellular construction then arranges that the local estimates, fine multiplicities, and coarse multiplicity all refer to one refinement.

6.1. The weighted profile inequality.

The coefficient $\mathfrak{P}_K$ compares the global body-mass-weighted shaded density with the sum of the parentwise responses $\alpha_W d_W^K$. Its denominator penalizes precisely those parents for which a large load is paired with a weak local efficiency.

Lemma 6.1 (Profile Hölder inequality)**.** *Let* $K > 1$ *and* $q = K/(K-1)$*. On a finite index set, suppose* $p_W > 0$*,* $\sum_W p_W = 1$*,* $D_W > 0$*,* $\alpha_W > 0$*, and* $d_W \ge 0$*. Then*

$$\sum_W p_W \alpha_W d_W^K \ge \mathfrak{P}_K(\mathbf{D}, \boldsymbol{\alpha}; \mathbf{p}) \left(\frac{\sum_W p_W D_W d_W}{\sum_W p_W D_W}\right)^K. \tag{61}$$

Moreover,

$$0 < \mathfrak{P}_K \le \max_W \alpha_W. \tag{62}$$

If $D_{\max}/D_{\min} \le \Lambda_D$ *and* $\alpha_W \ge \alpha_0$*, then*

$$\mathfrak{P}_K \ge \Lambda_D^{-K}\alpha_0. \tag{63}$$

If $D_W \equiv D_0$ *and* $\alpha_W \equiv \alpha_0$*, then* $\mathfrak{P}_K = \alpha_0$.

Proof. Write

$$\sum_W p_W D_W d_W = \sum_W p_W(\alpha_W^{1/K} d_W)(D_W \alpha_W^{-1/K}).$$

Hölder gives

$$\sum_W p_W D_W d_W \le \left(\sum_W p_W \alpha_W d_W^K\right)^{1/K} \left(\sum_W p_W D_W^q \alpha_W^{-1/(K-1)}\right)^{1/q}.$$

Rearranging gives (61).

Let $\alpha_* = \max_W \alpha_W$. Since $\alpha_W^{-1/(K-1)} \ge \alpha_*^{-1/(K-1)}$ and $\sum p_W D_W^q \ge (\sum p_W D_W)^q$, the denominator in (9) is at least $\alpha_*^{-1}(\sum p_W D_W)^K$, and hence (62); in particular $\mathfrak{P}_K \le 1$ when $\alpha_W \le 1$. Under the balanced hypotheses, the numerator is at least $D_{\min}^K$, while the denominator is at most $D_{\max}^K \alpha_0^{-1}$, proving (63). Direct substitution gives the constant-profile identity. □

Remark 6.2 (Equality and optimal scalar coefficient). Equality in (61) holds precisely, on the positive indices, when

$$\alpha_W d_W^{K-1} \text{ is proportional to } D_W,$$
$$d_W \text{ is proportional to } \left(\frac{D_W}{\alpha_W}\right)^{1/(K-1)}.$$

Thus $\mathfrak{P}_K$ is the optimal coefficient for this scalar Hölder assembly inequality. This optimality concerns the assembly step; the geometric argument may carry additional losses.

6.2. **A common cellular refinement.** We next isolate the global construction. The input is a family of parentwise local responses whose coefficients were fixed before refinement. Four mass-weighted restrictions are used: the first two regularize the inner and global fine multiplicities, the third regularizes the indexed number of active parents in a cell, and the fourth regularizes the fine mass of a cell. A null cleanup between the second and third restrictions makes positive-measure cell activity compatible with pointwise fibres. Since every later restriction is by a common union of whole half-open cells, it preserves the multiplicity information obtained earlier.

Let $\mathcal{V} = \dot{\bigcup}_{W\in\mathcal{W}} \mathcal{V}_W$ be finite indexed families of bounded bodies in $\mathbb{R}^4$, with $Y(V) \subset V \subset W$. Assume every parent group has positive initial shaded mass; initial zero-mass groups may be removed before invoking the theorem. Write

$$M_0 = \sum_{V\in\mathcal{V}} |Y(V)|, \qquad T_0 = \sum_{V\in\mathcal{V}} |V|, \qquad T_W = \sum_{V\in\mathcal{V}_W} |V|, \qquad D_W = \frac{T_W}{|W|}.$$

The T_W and D_W remain frozen through every refinement. Suppose the parent volumes and ordered John semiaxes are uniformly comparable to fixed scales $w_1 \le \cdots \le w_4$. Put

$$\ell = h = w_1/100, \qquad W^\sharp = N_{w_1}(W),$$

and assume $|W^\sharp| \asymp |W|$.

For each parent fix, before any refinement, a number $\alpha_W \in (0,1]$. Assume that for every measurable refinement and every parent with positive refined mass,

$$|N_h(U'_W)| \ge \Gamma \alpha_W d_W^K |W|, \qquad d_W = \frac{\sum_{V\in\mathcal{V}_W}|Y'(V)|}{T_W}, \qquad K > 1, \tag{64}$$

with common $\Gamma > 0$ and K. Let $N = \#\mathcal{V}$, $P = \#\mathcal{W}$, and

$$\mathcal{G} = 2 + \log(2+N) + \log(2+P) + \log(2 + w_4/w_1). \tag{65}$$

Lemma 6.3 (Grid-neighbor covering)**.** *Let $\mathcal{Q}_\ell$ be the half-open grid of side ℓ in $\mathbb{R}^d$, and let U be measurable such that the grid cube containing every point of U has positive-measure intersection with U. If*

$$Z_\ell(U) = \bigcup \{Q \in \mathcal{Q}_\ell : |U \cap Q| > 0\},$$

then

$$|N_h(U)| \le C_d(1 + h/\ell)^d |Z_\ell(U)|, \tag{66}$$

and $Z_\ell(U) \subset N_{\sqrt{d}\,\ell}(U)$.

Proof. If $y \in N_h(U)$, choose $x \in U$ with $|x - y| < 2h$. The lattice indices of their half-open grid cubes differ in each coordinate by at most $2 + \lceil 2h/\ell \rceil$. The cube of x is active, so the cube of y lies among at most $C_d(1 + h/\ell)^d$ neighbors of an active cube. Disjointness and equal grid volume prove the first assertion. If Q is active, every point of Q is within $\sqrt{d}\,\ell$ of a point of $U \cap Q$. □

Theorem 6.4 (Profile cellular factoring)**.** *Under the preceding hypotheses, there are a single refinement Y', coarse shadings $Z(W^\sharp)$ for every parent index, a nonempty positive final-mass set*

$$\mathcal{W}_+ = \{W : \ M'_W = \sum_{V\in\mathcal{V}_W} |Y'(V)| > 0\},$$

positive dyadic a, b, c, M, and $\sigma = M/c > 0$ such that $Z(W^\sharp) = \varnothing$ for $W \notin \mathcal{W}_+$ and

$$\sum_V |Y'(V)| \gtrsim \mathcal{G}^{-4} M_0. \tag{67}$$

The containment (13), the pointwise conclusions (15)–(17), the fibre equality (19), and local uniformity (20) hold with the full frozen coarse family $\mathcal{W}^\sharp$. For every $W \in \mathcal{W}_+$,

$$\mu(\mathcal{V}, Y) \lesssim \mathcal{G}^4 \mu(\mathcal{W}^\sharp, Z) \mu(\mathcal{V}_W, Y'), \tag{68}$$

and

$$\lambda(\mathcal{W}^\sharp, Z) \gtrsim \Gamma\mathcal{G}^{-4K}\mathfrak{P}_K(\mathbf{D}, \boldsymbol{\alpha}; \mathbf{p})\lambda(\mathcal{V}, Y)^K, \tag{69}$$

where

$$p_W = \frac{|W|}{\sum_{W'}|W'|}.$$

Proof. We construct the refinement through four nested mass-weighted selections. The first two act directly on the fine multiplicities. After removing null cell pieces, the third records how many parent indices are active in a cell, and the fourth makes the fine mass comparable across the retained cells. Each selection is charged to the current fine shaded mass, while the common restrictions preserve all uniformities already obtained.

Inner multiplicity. For the current shadings define

$$m_W(x) = \sum_{V \in \mathcal{V}_W} \mathbf{1}_{Y(V)}(x).$$

The identity $\sum_W \int m_W = \sum_V |Y(V)|$ and $1 \le m_W \le N$ on the fine union permit a dyadic choice of a carrying a logarithmic fraction of the fine mass. Intersecting every shading in a parent with $\{a \le m_W < 2a\}$ makes the inner multiplicity uniform on the surviving parent union.

Global fine multiplicity. For the refined shading let $m = \sum_W m_W$. Select one dyadic c by fine mass and intersect every shading with the common set $\{c \le m < 2c\}$. Because the same set is used for every child, the inner count obtained in the preceding step remains valid at every surviving point.

Positive-measure cell activity. Let $U_W^{(2)}$ be the parent union after these selections and partition $\mathbb{R}^4$ into half-open cubes Q of side ℓ. Remove from every child shading the same null set

$$E = \bigcup_{\substack{W,Q \\ |U_W^{(2)} \cap Q| = 0}} (U_W^{(2)} \cap Q). \tag{70}$$

Only finitely many cells meet the bounded parents, so E is null and the previous mass and multiplicity conclusions are unchanged. After removing it from every child, any surviving fine point makes its assigned parent active in its cell in the positive-measure sense required by the grid argument.

Indexed active-parent multiplicity. Call W active in Q if $|U_W^{(2)} \cap Q| > 0$, and let b_Q be the number of active parent indices. On every positive-fine-mass cell, $1 \le b_Q \le P$. Select a dyadic b by the fine cell masses

$$M_Q = \sum_V |Y(V) \cap Q|,$$

and retain the cells with $b \le b_Q < 2b$. This whole-cell restriction preserves the two fine multiplicity ranges and makes the indexed coarse count uniform on every retained cell.

Cell fine mass. Let $A = \sum_Q M_Q > 0$ after the third selection and let J be the number of cells with $M_Q > 0$. Discard cells with

$$M_Q < \frac{A}{2J}. \tag{71}$$

This loses less than $A/2$. The remaining positive masses have ratio at most $2J$, so one dyadic level

$$M \le M_Q < 2M \tag{72}$$

carries a logarithmic fraction of the mass. Intersect every child shading with the same union of selected whole cells. The final cell masses are therefore comparable without disturbing any of the three multiplicity ranges.

The number of cells satisfies

$$J \lesssim P \prod_{j=1}^{4} (1 + w_j/w_1). \tag{73}$$

Indeed, a cube meeting W lies in $N_{2\ell}(W)$, and disjointness of the grid and the John-box estimate give

$$\#\{Q : Q \cap W \neq \varnothing\}\ell^4 \le |N_{2\ell}(W)| \lesssim \prod_{j=1}^{4} (w_j + \ell).$$

Thus each of the four selections costs at most a factor controlled by $\mathcal{G}$, which gives (67). The construction has used the pointwise and cellular quantities, while retaining the complete range of parent loads for the profile assembly.

Let $\mathcal{W}_+$ be the parents with positive final mass. For every parent, including those outside $\mathcal{W}_+$, define

$$Z(W^\sharp) = \bigcup \{Q : Q \text{ is finally selected and } |U'_W \cap Q| > 0\}. \tag{74}$$

This is empty for $W \notin \mathcal{W}_+$. If a parent was active in a finally selected cell, the whole-cell restriction preserves its positive-measure fine piece, so it belongs to $\mathcal{W}_+$. Consequently the coarse multiplicity on a selected cell is the previously selected $b_Q \in [b, 2b)$. Parents outside $\mathcal{W}_+$ contribute empty coarse shadings but remain in the fixed denominator. Since a grid cube has diameter $2\ell < w_1$,

$$U'_W \subset Z(W^\sharp) \subset N_{w_1}(U'_W) \cap W^\sharp.$$

The common whole-cell restrictions preserve the inner and global counts.

At a final fine point, let $p(x)$ count assigned parent indices contributing a child. Then

$$c \le m_{\mathcal{V},Y'}(x) < 2a\, p(x), \qquad p(x) \le b_Q < 2b,$$

and hence

$$c < 4ab. \tag{75}$$

For $W \in \mathcal{W}_+$ and for the full coarse family,

$$a \le \mu(\mathcal{V}_W, Y') < 2a, \qquad b \le \mu(\mathcal{W}^\sharp, Z) < 2b.$$

Writing $M' = \sum_V |Y'(V)|$ and $U' = U(\mathcal{V}, Y')$, we have

$$\mu(\mathcal{V}, Y) \le \frac{M_0}{|U'|} \lesssim \mathcal{G}^4 \frac{M'}{|U'|} < 2\mathcal{G}^4 c.$$

Together with (75), this proves (68).

Unique half-open cell ownership, the global null cleanup, and the disjoint assigned index union give

$$(\mathcal{V})_{Y'}(x) = \dot{\bigcup}_{\substack{W \in \mathcal{W} \\ x \in Z(W^\sharp)}} (\mathcal{V}_W)_{Y'}(x).$$

On a selected cell,

$$c|U' \cap Q| \le M_Q < 2c|U' \cap Q|, \qquad M \le M_Q < 2M,$$

so $|U' \cap Q| \asymp \sigma = M/c$. A ball of radius w_1 centered in the coarse union contains its center cell and meets only $O(1)$ grid cells, proving local uniformity.

For the density conclusion, the null cleanup gives the pointwise hypothesis of Lemma 6.3, and $Z(W^\sharp) = Z_\ell(U'_W)$. Since $h = \ell$,

$$|Z(W^\sharp)| \gtrsim |N_h(U'_W)|. \tag{76}$$

Put $d_W = 0$ for $W \notin \mathcal{W}_+$. The local input, summation, and $|W^\sharp| \asymp |W|$ give

$$\lambda(\mathcal{W}^\sharp, Z) \gtrsim \Gamma \sum_W p_W \alpha_W d_W^K. \tag{77}$$

Moreover,

$$\sum_W p_W D_W d_W = \frac{M'}{\sum_W |W|}, \qquad \sum_W p_W D_W = \frac{T_0}{\sum_W |W|}.$$

Thus

$$\frac{\sum_W p_W D_W d_W}{\sum_W p_W D_W} = \frac{M'}{T_0} \gtrsim \mathcal{G}^{-4} \lambda(\mathcal{V}, Y). \tag{78}$$

Apply Lemma 6.1 to (77) and use (78), which yields (69); the four-selection loss is raised to K. □

6.3. **From the local response to the main theorem.** It remains to match the local geometric response from Section 5 with the abstract cellular theorem. The initial reduction to $\mathcal{W}_0$ provides the positive-mass hypothesis, parent comparability makes the local constants uniform, and the fixed local efficiencies become the coefficients in the Hölder profile.

Proof of Theorem 2.2. Restrict first to the parent groups in $\mathcal{W}_0$ from (10). Every discarded input shading is null; finiteness therefore preserves both the shaded mass and the measure of the fine union. Let $\mathcal{V}_0$ be the remaining assigned child family and set

$$\lambda_0 = \frac{\sum_{V \in \mathcal{V}_0} |Y(V)|}{\sum_{V \in \mathcal{V}_0} |V|}.$$

Then $\lambda_0 \ge \lambda$.

Fix $\ell = h = w_1/100$. For every $W \in \mathcal{W}_0$,

$$\frac{h}{a_1(W)} \in \left[\frac{1}{100\Lambda_W}, \frac{\Lambda_W}{100}\right], \qquad \frac{a_1(W)}{a_4(W)} \asymp_{\Lambda_W} \frac{w_1}{w_4}.$$

The local logarithm is bounded by $\mathfrak{L}$, uniformly in W. Corollary 5.7, in its intrinsic form, supplies (64) for every later measurable refinement with

$$\Gamma \gtrsim_{\varepsilon,\Lambda_W} \mathfrak{L}^{-A_{0,\varepsilon}} \left(\frac{w_1}{w_4}\right)^{\varepsilon}, \qquad \alpha_W = \kappa_W^{(K_\varepsilon)}, \qquad K_\varepsilon \geq 2.$$

The coefficient $\alpha_W \in (0,1]$ is determined before the cellular refinement. Lemma 5.2 gives $|W^\sharp| \asymp_{\Lambda_W} |W|$ and the required Katz–Tao stability.

Apply Theorem 6.4 to $\mathcal{W}_0$ and $\mathcal{V}_0$, and extend its output to the original indexed family by setting $Y'(V) = \varnothing$ for $V \in \mathcal{V} \setminus \mathcal{V}_0$. The extension is a refinement. Since the discarded input shadings form a finite null union, $\mu(\mathcal{V}, Y) = \mu(\mathcal{V}_0, Y)$; the empty outputs also remove those indices from every refined pointwise fibre. With $\alpha_W = \kappa_W^{(K_\varepsilon)}$, the cellular profile is (11). Since $\lambda_0^{K_\varepsilon} \geq \lambda^{K_\varepsilon}$, its density estimate proves (14). Its mass, compatibility, multiplicity, product, fibre, and local-uniformity conclusions prove (12)–(20) after increasing one A_ε to dominate the local and four-selection logarithmic powers. Finally,

$$\Delta_{\max}(\mathcal{W}_0^\sharp) \lesssim_{\Lambda_W} \Delta_{\max}(\mathcal{W}_0) \leq \Delta_{\max}(\mathcal{W}),$$

which gives (21) and completes the proof. □

Proof of Corollary 2.3. Lemma 5.4 and (7)–(8) give, parentwise,

$$\kappa_W^{(K_\varepsilon)} \geq C^{-1} \eta_W^{(K_\varepsilon)}.$$

Substitution into the denominators of (11) and (22) yields

$$\mathfrak{P}_\varepsilon^{\mathrm{int}} \geq C^{-1} \mathfrak{P}_\varepsilon^{\mathrm{elem}}.$$

Theorem 2.2 now gives (23) with the same refinement and all the same structural conclusions. □

7. Consequences and the comparable-scale regime

The profile theorem becomes more familiar when the parent loads and local inflation efficiencies are balanced. We first record this direct consequence and use a mixed geometry example to show that inflation balance is weaker than coordinatewise axis comparability. We then recover the comparable-scale conclusion by selecting a load level before applying the main theorem.

7.1. **Balanced profiles and mixed child geometry.**

Corollary 7.1 (Balanced elementary profile)**.** *In Corollary 2.3, suppose*

$$\frac{\max_{W\in\mathcal{W}_0} D_W}{\min_{W\in\mathcal{W}_0} D_W} \leq \Lambda_D, \qquad \eta_W^{(K_\varepsilon)} \geq \eta_0 > 0 \quad (W \in \mathcal{W}_0).$$

Then the same refinement satisfies

$$\lambda(\mathcal{W}_0^\sharp, Z) \gtrsim_{\varepsilon,\Lambda_W} \mathfrak{L}^{-A_\varepsilon} \left(\frac{w_1}{w_4}\right)^{\varepsilon} C^{-1}\Lambda_D^{-K_\varepsilon}\eta_0\lambda^{K_\varepsilon}. \tag{79}$$

Proof. Apply (63) to $\alpha_W = \eta_W^{(K_\varepsilon)}$ in (22). □

The parallel-volume estimate (45) gives the useful criterion

$$r_W(V) \asymp \prod_{j=1}^{4}\left(1+\frac{h}{a_j(V)}\right). \tag{80}$$

Thus, if the products on the right are uniformly comparable among the children in a fixed parent, then $r_W^- \asymp \overline{r}_W \asymp r_W^+$ and $\eta_W^{(K_\varepsilon)} \asymp 1$. This product condition allows coordinatewise axis ratios to be unbounded.

Example 7.2 (Mixed sub-buffer shapes)**.** Let $\tau \ll h$ and take equal-volume convex children whose John-axis quadruples are

$$(\tau^3,\tau,\tau,\tau) \quad \text{or} \quad (\tau^2,\tau^2,\tau,\tau).$$

Both products of the four axes equal τ^6, whereas the first two coordinatewise axis ratios between the two shapes are τ and τ^{-1}. Hence no fixed coordinatewise comparability constant survives as $\tau \to 0$. Since all axes are $o(h)$,

$$|N_h(V)| \asymp h^4, \qquad r_W(V) \asymp \frac{h^4}{|V|}.$$

Equal child volumes therefore make all inflation ratios comparable and give $\eta_W^{(K_\varepsilon)} \asymp 1$. Whenever the parent loads are also balanced, Corollary 7.1 gives a nondegenerate conclusion although no uniform coordinatewise child-axis comparison is available.

7.2. **The comparable-scale specialization.**

Corollary 7.3 (Comparable-scale buffered factoring)**.** *Assume the hypotheses of Corollary 2.3 and, in addition, fix $\Lambda_V \ge 1$ such that for every pair of child indices V, V' and every $1 \le j \le 4$,*

$$\Lambda_V^{-1} a_j(V') \le a_j(V) \le \Lambda_V a_j(V').$$

Then there is a nonempty selected parent subfamily $\mathcal{W}' \subset \mathcal{W}$ for which all the structural conclusions of Theorem 2.2 hold, every retained coarse index has positive final mass, and

$$\lambda(\mathcal{W}'^\sharp, Z) \gtrsim_{\varepsilon,\Lambda_V,\Lambda_W} \mathfrak{L}^{-A_\varepsilon} \left(\frac{w_1}{w_4}\right)^{\varepsilon} C^{-1}\lambda^{K_\varepsilon}. \tag{81}$$

In particular, the conclusion includes the one-refinement, multiplicity-product, fibre, local-uniformity, and Katz–Tao properties of Theorem 2.2, now on the selected parent family.

Proof. Discard first the initial zero-shaded-mass parent groups. This preserves the numerator of the fine density and decreases its denominator. Coordinatewise child-axis comparability and the John-volume estimate imply that all child volumes are comparable to one value v. Parent-axis comparability makes all parent volumes comparable as well. If $n_W = \#\mathcal{V}_W$, then

$$D_W = \frac{T_W}{|W|} \asymp_{\Lambda_V,\Lambda_W} n_W \frac{v}{|W|}. \tag{82}$$

For a nonempty group, $1 \le n_W \le N = \#\mathcal{V}$, so the positive loads occupy $O_{\Lambda_V,\Lambda_W}(1+\log(2+N))$ dyadic levels.

Choose a load level $D \le D_W < 2D$ carrying a reciprocal-logarithmic fraction of the original shaded mass, and call its parent family $\mathcal{W}_1$. If M_1, T_1 are the selected shaded and body masses, let $\mathcal{V}_1 = \dot{\bigcup}_{W\in\mathcal{W}_1}\mathcal{V}_W$ and $U_1 = U(\mathcal{V}_1, Y)$ and $U_{\text{full}} = U(\mathcal{V}, Y)$. Then

$$M_1 \gtrsim \mathfrak{L}^{-1} M_0, \qquad \lambda_1 = \frac{M_1}{T_1} \ge \frac{M_1}{T_0} \gtrsim \mathfrak{L}^{-1}\lambda.$$

The selected loads are 2-comparable. Since $U_1 \subset U_{\text{full}}$, the retained-mass estimate also compares the original and selected average multiplicities:

$$\mu(\mathcal{V}, Y) = \frac{M_0}{|U_{\text{full}}|} \lesssim \mathfrak{L}\frac{M_1}{|U_{\text{full}}|} \le \mathfrak{L}\frac{M_1}{|U_1|} = \mathfrak{L}\mu(\mathcal{V}_1, Y).$$

At the common radius $h = w_1/100$, (80) and the global coordinatewise child-axis hypothesis imply

$$r_W^- \asymp_{\Lambda_V} \overline{r}_W \asymp_{\Lambda_V} r_W^+, \qquad \eta_W^{(K_\varepsilon)} \gtrsim_{\Lambda_V,K_\varepsilon} 1$$

for every $W \in \mathcal{W}_1$. Lemma 6.1, or Corollary 7.1, therefore bounds the selected elementary profile below by a positive constant depending only on $\Lambda_V, \Lambda_W, \varepsilon$.

Apply Corollary 2.3 to the selected family $(\mathcal{W}_1, \mathcal{V}_1)$, and extend its output by $Y'(V) = \varnothing$ for $V \in \mathcal{V} \setminus \mathcal{V}_1$. Its mass conclusion, combined with $M_1 \gtrsim \mathfrak{L}^{-1}M_0$, gives the mass bound relative to the original family after increasing A_ε. Its density conclusion contains $\lambda_1^{K_\varepsilon} \gtrsim \mathfrak{L}^{-K_\varepsilon}\lambda^{K_\varepsilon}$, and hence yields (81) after the same increase. The displayed comparison of average multiplicities and the selected-family product give the full-family multiplicity product with one further factor $\mathfrak{L}$, which is absorbed by the same increase of A_ε. The empty extension makes all full-family pointwise multiplicity and fibre statements literal.

The profile theorem keeps final zero-mass parent buffers in the denominator fixed by the selected input family. Delete precisely those empty-shading coarse indices. The coarse numerator, coarse union, all pointwise multiplicities, and fibre equality are unchanged; coarse density can only increase and $\Delta_{\max}$ can only decrease. The remaining parent family $\mathcal{W}'$ consists precisely of positive final-mass parents, so every average multiplicity in the corollary is defined. All structural conclusions are therefore inherited by $\mathcal{W}'$. □

Remark 7.4. Corollary 7.3 normalizes the coarse family after selecting a parent-load level. Theorem 2.2, by contrast, keeps the entire initial positive-mass support and records load imbalance through its profile. The two conclusions agree in the balanced regime but use different denominators before that specialization.

8. Structural limitations and model examples

The definitions in the main theorem reflect several distinct geometric and normalization constraints. The examples below explain them one at a time: the external buffer, indexed counting, mean-normalized inflation, the parent-load profile, the projection direction, and the two cellular discretizations. Each model addresses a particular mechanism rather than asserting global optimality of the full theorem.

8.1. **Why the coarse shading lives in the external buffer.** Fix $L \gg r > 0$ and, in $\mathbb{R} \times \mathbb{R}^3$, consider the cone

$$W = \{(t, y) : 0 \le t \le L,\ |y| \le (r/L)t\}. \tag{83}$$

Its shortest John scale is comparable to r. At the apex,

$$|B(0,r) \cap W| \asymp \int_0^r \left(\frac{rt}{L}\right)^3 dt \asymp \frac{r^7}{L^3}, \qquad \frac{|B(0,r) \cap W|}{|B(0,r)|} \asymp (r/L)^3. \tag{84}$$

For the lower bound one may integrate only over $r/4 \le t \le r/2$; the upper bound uses the cone cross-section directly. Let $U_\eta = W \cap B(0,\eta)$ with $0 < \eta \ll r$. Because the apex belongs to U_η, the full neighborhood $N_r(U_\eta)$ contains $B(0,r)$. In contrast,

$$W \cap N_r(U_\eta) \subset W \cap B(0, r+\eta),$$

and the same integration gives an $O((r/L)^3)$ fraction of an r-ball. Thus no aspect-independent comparison can replace the coarse shading in the external buffer $N_r(W)$ by $W \cap N_r(U_\eta)$. Lemma 5.2 provides the alternative used in Theorem 2.2: the external buffer has comparable parent volume and stable Katz–Tao density.

8.2. **Indexed multiplicity under coincident geometry.** Let there be R different parent indices with identical geometric parent W, and assign to each one a single identical child and identical nonnull shading. The global fine multiplicity is R, the inner multiplicity in every assigned group is 1, and the indexed coarse multiplicity is R. Hence the multiplicity product is exact up to constants. If the coarse parents were geometrically deduplicated without weights, the coarse multiplicity would become 1 and the product would fail by the arbitrary factor R. This example forces indexed counting in (17), (18), and (21).

The same mechanism appears in Theorem 4.6. If $Q_i = Q$ and $E_i = E$ for all N indices, then the indexed Katz–Tao density is $m = N$, whose inverse cancels the indexed body mass $N|Q|$. This balance is the reason the passage

from the essentially-distinct theorem to the indexed theorem is carried out inside the paper.

8.3. **Mean-normalized density conversion.** Let $W = [0,1]^4$, fix $h = c_0 a_1(W)$, include $V_0 = W$, and partition W into $N = m^4$ congruent cubes $V_1, \dots, V_N$ of side m^{-1}. The indexed family has bounded relative Frostman constant: its total body density in W is 2, a proper convex test body cannot contain V_0, and the disjoint small cubes contained in such a test body have total volume at most the test volume.

For large m,

$$r_0 \asymp 1, \qquad r_j \asymp N \quad (1 \le j \le N).$$

Since the large child and all small children each contribute total body mass 1,

$$r_W^- \asymp 1, \qquad \overline{r}_W = \frac{r_0|W| + \sum_{j=1}^N r_j|V_j|}{2|W|} \asymp N, \qquad r_W^+ \asymp N.$$

Shade only V_0. Then

$$d_W = \frac{1}{2}, \qquad d_{W,h}^{\mathrm{wt}} = \frac{r_0|W|}{r_0|W| + \sum_{j=1}^N r_j|V_j|} \asymp N^{-1} \asymp \frac{r_W^-}{\overline{r}_W} d_W.$$

Thus (5) is sharp up to constants. The elementary local efficiency is

$$\eta_W^{(K)} = \frac{\overline{r}_W}{r_W^+} \left(\frac{r_W^-}{\overline{r}_W} \right)^K \asymp N^{-K}.$$

Thus the example saturates the mean-normalized conversion without producing an additional outer inflation-ratio factor.

8.4. **Load imbalance and the profile coefficient.** Take P pairwise disjoint congruent parents with probability weights $p_j = 1/P$. Let $D_1 = R$, $D_j = 1$ for $j \ge 2$, and $\alpha_j = 1$. Concentrate the fine shaded density on the first parent. When $R \gg P$, the global fine density can be close to 1, but a coarse denominator frozen over all P parents has density at most $O(P^{-1})$ if only the first parent carries substantial coarse shading. Algebraically,

$$\mathfrak{P}_K = \frac{((R+P-1)/P)^K}{((R^{K/(K-1)}+P-1)/P)^{K-1}} \asymp P^{-1} \qquad (R \gg P).$$

The displayed profile therefore decays like P^{-1} in this regime. The comparable-scale corollary first selects one load level and consequently uses a different coarse normalization. By Remark 6.2, the profile is optimal for the scalar Hölder step; this example does not address optimality of the remaining geometric losses.

8.5. **Choice of projection direction.** Let $V = W$ be a box with side lengths $w_1 \ll w_2 \le w_3 \ll w_4$, take $h = w_1$, and project along the longest direction. Then

$$\frac{|N_h(V)|}{|\pi N_h(V)|} \asymp w_4,$$

not h. The upper estimate in Lemma 5.3 contains the width in the projection direction. Choosing a shortest John direction turns that width into $O(h)$ while still permitting arbitrarily tilted children.

8.6. **Affine orientation and cell-mass selection.** The affine parameterization in Section 4 is needed even when all boxes have the same center and long axis. For

$$R_\phi = \mathrm{Rot}_{e_3}(\phi)([-\delta,\delta]^2 \times [-1,1]), \qquad \phi = j\delta, \quad 0 \le \phi \le \pi/4,$$

all boxes have the same center and long axis, and any two square cross-sections overlap in more than half their area. The non-essential-distinctness graph contains a clique of size comparable to δ^{-1}. Lemma 4.4 avoids this aspect loss by measuring relative affine transforms modulo the finite symmetry group of the box.

Finally, arbitrary measurable shadings can place masses 2^{-2^k} in successive grid cells. Their positive dyadic levels are not bounded by a logarithm of the family cardinality or aspect ratio. The relative cutoff (71) removes less than half the current mass and leaves a mass ratio at most $2J$; this is why the fourth profile selection costs $O(\log(2+J))$ and remains valid for densities smaller than every fixed scale power.

Finally, the product construction in Proposition 3.2 shows that the codimension-one lift preserves, rather than improves, both the factor H and the density response inherited from the lower-dimensional theorem.

Appendix A. Dependence of constants and logarithmic losses

Theorem 2.2 collects several logarithmic factors into one exponent A_ε. We record their origins here and state the uniformity of the remaining constants.

A.1. **Local analytic losses.** For a local assigned family of cardinality N_W, the indexed collision reduction contributes

$$L_{N_W}^{-1} = (1 + \lceil \log_2 N_W \rceil)^{-1}.$$

The three ordered projected John axes range between h and R, so their dyadic binning contributes

$$J_h^{-1} \gtrsim (1 + \log_+(R/h))^{-3}.$$

Since $N_W \le \#\mathcal{V}$ and, uniformly in W, $R/h \asymp_{\Lambda_W} w_4/w_1$, their product is bounded below by a fixed negative power of

$$\mathfrak{L} = 2 + \log(2+\#\mathcal{V}) + \log(2+\#\mathcal{W}) + \log(2+w_4/w_1).$$

Thus the cardinality and axis-binning losses remain in $\mathfrak{L}$, separately from the power $(w_1/w_4)^\varepsilon$.

A.2. **Cellular and profile losses.** The four mass-weighted selections in the cellular theorem have at most

$$O(\log(2+\#\mathcal{V})),\quad O(\log(2+\#\mathcal{V})),\quad O(\log(2+\#\mathcal{W})),\quad O(\log(2+J))$$

choices, respectively. They leave all parent loads in the Hölder profile. The cell count satisfies

$$J \lesssim \#\mathcal{W}\prod_{j=1}^{4}(1+w_j/w_1),$$

and therefore

$$\log(2+J) \lesssim 1+\log(2+\#\mathcal{W})+4\log(2+w_4/w_1) \lesssim \mathfrak{L}.$$

The retained mass is thus $\gtrsim \mathfrak{L}^{-4}$ times the incoming mass. The profile Hölder normalization raises this fraction to K_ε, giving $\mathfrak{L}^{-4K_\varepsilon}$ in the coarse density. The comparable-scale specialization uses one additional parent-load selection. One admissible exponent covering both the profile theorem and that specialization is

$$A_\varepsilon \geq A_{0,\varepsilon}+5K_\varepsilon+6.$$

For the profile theorem itself, the four cellular selections account for the global selection loss.

A.3. **Base-family profile quantities.** For each parent in the initial positive-mass support, the following quantities are determined before the cellular refinement:

$$T_W,\quad D_W,\quad r_W^-,\quad \overline{r}_W,\quad r_W^+,\quad C_{F,h}^{\mathrm{wt}},\quad \kappa_W,\quad \eta_W.$$

The exponent $K_\varepsilon \geq 2$ comes from the three-dimensional response and is unchanged by the lift. The parent probabilities p_W, the profile $\mathfrak{P}_\varepsilon$, and the fine density use the denominators fixed in Section 2. The data profile therefore remains displayed separately from $\mathfrak{L}^{-A_\varepsilon}$ and the implicit constant.

A.4. **Uniformity in the nominal parent scale.** The proof uses the common analytic radius $h = w_1/100$, whereas Theorem 5.6 is written with $h = c_0 a_1(W)$. The parent comparison hypothesis gives

$$c_0 = \frac{h}{a_1(W)} \in \left[\frac{1}{100\Lambda_W}, \frac{\Lambda_W}{100}\right].$$

Every geometric constant in Lemmas 5.3–5.5 is uniform on this compact interval. Also

$$\frac{a_1(W)}{a_4(W)} \asymp_{\Lambda_W} \frac{w_1}{w_4}, \qquad w_1 \leq \Lambda_W a_1(W),$$

which justifies both the common aspect factor and the common buffer radius.

The profile theorem's implicit constants may depend on ε, the dimension, and Λ_W; the comparable-scale specialization may additionally depend on Λ_V. They are uniform in λ, the family cardinalities, duplicate multiplicity, child orientation, child aspect ratio, and child minimum scale. The explicit profiles and logarithms retain the corresponding base-family data.

Appendix B. Derivation of the three-dimensional input

Proposition 4.1 combines several statements from Wang and Zahl with a short localization and density-threshold argument. This appendix gives the notation dictionary for that derivation and distinguishes the source inputs from the indexed extension proved in Section 4.

a, b *(manuscript)* $\longleftrightarrow$ a, b *(source).* These are the normalized John dimensions in Definitions 3.1 and 5.4; here $a = q_1/q_3$ and $b = q_2/q_3$.

$N, v \longleftrightarrow \#\mathcal{P}, |P|$. These are the cardinality and comparable body-volume scale in (5.5), with $v \asymp ab$ after normalizing the longest dimension to one.

$\rho \longleftrightarrow$ *the dense-shading parameter.* Definition 3.1 uses the same average-mass notion, $\sum_P |Y(P)| \geq \rho \sum_P |P|$, and Definition 5.4 assumes $\rho \geq a^\eta$.

$m = \Delta_{\max}(\mathcal{Q}) \longleftrightarrow C_{\mathrm{KT-CW}}(\mathcal{P})$. The source locations are Definition 1.3′(A), equation (4.1), and Remark 4.2(A). The quantities agree after identifying $\mathcal{P} = \mathcal{Q}$; the source family at this stage is a set, not our later indexed multiset.

$\ell \longleftrightarrow C_{\mathrm{F-SW}}(\mathcal{P})$. The source locations are Definition 1.3′(B), equation (4.2), and Remark 4.2(A), with the test collection restricted to slabs. This auxiliary slab constant is not the relative convex Frostman constant $C_F(\mathcal{Q}, B')$ used later in the paper.

$\mathcal{D}(\mathcal{Q}) \longleftrightarrow D(\mathcal{P})$. The source locations are Definition 5.4, equation (5.6), and Remark 5.6. The calligraphic letter here avoids collision with the manuscript's other density symbols.

$t, t \longleftrightarrow \sigma, \omega$. These are the positive parameters in $F(\sigma, \omega)$, Definition 5.4. In Proposition 4.1 both are chosen as $t = \varepsilon/10$, while the source's displayed a^ε loss is invoked with $\varepsilon = \varepsilon/2$; thus the source threshold exponent is $\eta_{\mathrm{WZ}}(\varepsilon/2; t, t)$.

With this dictionary, (5.5) agrees with (31) up to the fixed John comparability constants and the source constant κ. The facts $Nv \lesssim m$, $m \gtrsim 1$, and $\ell \lesssim a^{-1}$ come, respectively, from the fixed ambient localization, testing the Katz–Tao constant on one member, and the elementary slab-volume test. For every P, subfamily monotonicity gives $C_{\mathrm{KT-CW}}(\mathcal{P}[N_b(P)]) \leq m$, so Remark 5.6, Situation 2, gives $\mathcal{D} \lesssim m^{1/2}$. Hence, writing $B = m^{-3/2}\ell N v^{1/2}$,

$$B\mathcal{D} \lesssim m^{-1}\ell N v^{1/2} \lesssim \ell v^{-1/2} \lesssim a^{-2}.$$

Thus the positive source parameters are absorbed into a^ε by the derivation following (31), yielding (30). The branch $0 < \rho < a^\eta$ uses the largest shading and is internal to Proposition 4.1. The source chain begins at $D(0, 0)$, but the convex assertion used in the union estimate is $F(t, t)$ with $t = \varepsilon/10 > 0$.

The relevant source result chain is precise. Theorem 1.9 gives $D(0, 0)$. At $\sigma = \omega = t$, Definition 1.5 makes $D(t, t)$ weaker by the factor $(\#\mathcal{T})^{-t} \leq 1$. Proposition 1.6 gives $E(t, t)$, and Proposition 5.14 gives $F(t, t)$ for $0 < t \leq 2/3$ [10]. Remark 5.6, Situation 2, supplies the separate bound for the geometric factor D used above.

(1) Wang and Zahl's three-dimensional convex assertion $F(\sigma,\omega)$ is used for positive σ and ω, together with their endpoint theorem and equivalence result [10]. The source assertion assumes an essentially-distinct family and a positive density threshold. Proposition 4.1 converts it to the required all-density form by positive-parameter absorption and the separate largest-shading argument.
(2) The source definition of essential distinctness is (29), so coincident indexed bodies lie outside its scope. Lemma 4.4 and Theorem 4.6 provide the indexed extension, with the explicit loss L_N^{-1}.
(3) Guth, Wang, and Zahl formulate the three-dimensional factoring architecture and develop its induced-shading density mechanism [5]. Subsequent analysis of the printed non-cellular transitions gives a conditional cellular replacement for the relevant compatibility interface [6]. These works motivate the factoring architecture; the four-dimensional theorem uses the self-contained cellular argument of Section 6.
(4) The sticky works of Wang–Zahl and Choudhuri contain compatible coarse/fine refinements and multiplicity products [11, 2]. They provide context for the coarse/fine organization, while Theorem 2.2 is proved without their extremal or sticky hypotheses.
(5) The four-dimensional maximal theorem of Borges, Chan, Chen, Liu, Xi, and Zhan [1] locates the structural result within the broader four-dimensional problem. The factoring proof itself is independent of its restriction, two-ends, plany, and transversality conclusions.

The division of roles is therefore explicit: $F(t,t)$ is used at positive parameters, the small-density branch is proved in Proposition 4.1, and the passage from essentially-distinct bodies to indexed families is carried out by Theorem 4.6.

Appendix C. Supplementary boundary cases

The main text emphasizes the mechanisms that drive the estimates. The examples here clarify three set-theoretic and normalization conventions used by the common refinement.

C.1. **Coarse shadings and fine fibres.** Let a parent have positive fine shading in a small measurable subset of a selected cell Q. By definition, $Z(W^\sharp)$ contains the whole cell. At a point $x \in Q$ outside the fine shading one has $x \in Z(W^\sharp)$ but $(\mathcal{V}_W)_{Y'}(x) = \varnothing$. Thus (19) is valid because empty fibres contribute nothing, but the stronger parent-set identity

$$\{W : (\mathcal{V}_W)_{Y'}(x) \neq \varnothing\} = \{W : x \in Z(W^\sharp)\}$$

is false in general.

C.2. **The global null cleanup.** Start with several parents having positive shading in each of several cells. Add one parent with positive shading in one such cell and with nonempty null dust in the other cells. Positive-measure activity counts the added parent only in its positive cell, but the set-theoretic neighborhood of its union sees the dust in all cells. Without deleting the dust, neither the pointwise implication from fine contribution to active parent nor the grid-neighbor comparison (76) is valid everywhere.

Deleting null pieces separately for each parent is insufficient at points shared by different parents, because it can change the global count while leaving the point in the global union. The single set E in (70) is a finite union of null parent–cell pieces and is removed from every child. It preserves every mass and all previously selected pointwise multiplicities.

C.3. **Limits of unweighted Frostman inheritance.** Let $W = [0,1]^4$, include $V_0 = W$, and let

$$V_j = I_j \times [0,1]^3, \qquad 1 \le j \le N,$$

where the consecutive disjoint intervals I_j have length $\eta = N^{-4}$. The original indexed family has bounded relative Frostman constant: a proper convex test body cannot contain V_0, while the small slabs it contains have disjoint interiors and total volume at most that of their convex hull.

Take $h = N^{-2}$ and $P_j = N_h(V_j)$. The convex body

$$K = N_h([0, N\eta] \times [0,1]^3)$$

contains $P_1, \dots, P_N$ but not P_0. Since $N\eta \ll h$,

$$|K| \asymp h, \qquad |P_j| \asymp h, \qquad \frac{\sum_{j=1}^N |P_j|}{|K|} \asymp N.$$

Meanwhile the total thickened-body density in a fixed containing box is bounded. Thus the relative Frostman constant after thickening can grow like N, showing that an inflation-sensitive coefficient is unavoidable; that geometric calculation alone does not identify the sharp universal ratio.

The intrinsic quantity for the transfer is the exact weighted coefficient $C^{\mathrm{wt}}_{F,h}$ from (6). The universal comparison

$$C^{\mathrm{wt}}_{F,h} \le C\, r_+/\overline{r}$$

is the proof-valid upper bound of Lemma 5.4. Here the relevant scales are

$$r_0 \asymp 1, \qquad r_j \asymp h/\eta = N^2, \qquad r^- \asymp \overline{r} \asymp 1, \qquad r^+ \asymp N^2.$$

Indeed, the small slabs have total weighted mass $\sum_{j=1}^N r_j |V_j| \asymp Nh = N^{-1}$, whereas the large child contributes mass comparable to 1. Testing the weighted coefficient on the unthickened convex strip $[0, N\eta] \times [0,1]^3$ gives weighted density comparable to N^2. This example isolates the need for an inflation-sensitive coefficient; it does not assert sharpness of the full inheritance or profile theorem. Its radius is not tied to the shortest scale of

the unit parent. The scale-compatible construction in Section 8 separately treats the mean-normalized density conversion when $h \asymp a_1(W)$.

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